Order-Optimal Non-Adaptive One-Bit Mean Estimation in General Moment Classes

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1 Theoretical Setup

1.1 Problem setting

For a probability law D on $\mathbb{R}$, write $\mu(D) = \mathbb{E}_D X$. For fixed $k > 1$ and scales $\lambda \geq \sigma > 0$, define the unrestricted central-moment class

$$\mathcal{D}(k, \lambda, \sigma) := \left\{ D : \ \mu(D) \in [-\lambda, \lambda], \quad \mathbb{E}_D |X - \mu(D)|^k \leq \sigma^k \right\}.$$

No support, density, symmetry, likelihood, or tail-shape condition is imposed. All logarithms are natural. For $u, v > 0$, the notation $u \lesssim_k v$ means $u \leq C(k)v$ for a finite constant depending only on k, and $u \asymp_k v$ means both inequalities.

The observations are encoded one at a time into membership bits. A query is a Borel set $A \subseteq \mathbb{R}$, and the corresponding message is $\mathbf{1}\{X \in A\}$. We seek a fixed-horizon protocol in which all queries and all public randomness are committed before any message is seen. The performance criterion is unconditional absolute-error probability, jointly over the observations and every source of public protocol randomness.

1.2 Assumptions

Assumption 1 (Parameter domain)**.** *The number $k > 1$ is fixed and known, the scales $\lambda \geq \sigma > 0$ are known, $\delta \in (0, 1/2)$, and $0 < \epsilon \leq c_k \sigma$, where the positive constant $c_k < 1$ is chosen only as a function of k.*

Assumption 2 (Unrestricted central-moment class)**.** *The common law satisfies $D \in \mathcal{D}(k, \lambda, \sigma)$, with no stronger support, symmetry, density, or tail condition.*

Assumption 3 (Independent observations and seeds)**.** *All observations used by the localization and refinement blocks are independent and identically distributed with law D. The localization seed and all refinement levels, colors, branches, types, masks, and dithers are mutually independent and independent of every observation.*

Assumption 4 (Precommitted protocol)**.** *The sample split, localization seed, refinement variables, countable mask families, dithers, and median-of-means groups are sampled or fixed before any bit is observed, and all public randomness is available to the decoder. Every query set is Borel measurable and cannot depend on an earlier message. A localization output may be used only after the complete transcript has been collected, and then only in decoder-side arithmetic.*

1.3 Objective

The goal is a constructive non-adaptive protocol which, uniformly over $\mathcal{D}(k, \lambda, \sigma)$, uses one bit per independent observation and achieves error at most ϵ with failure probability at most δ. The horizon must have the exact three-regime dependence defined in Section 2, including the single logarithmic loss when $k = 2$ and no hidden dependence outside constants determined by the fixed k.

2 Preliminaries

The rate appearing in the main theorem is

$$r_k(\lambda, \sigma, \epsilon, \delta) := \log \frac{\lambda}{\sigma} + \begin{cases} \dfrac{\sigma^2}{\epsilon^2} \log \dfrac{1}{\delta}, & k > 2, \\ \dfrac{\sigma^2}{\epsilon^2} \log \dfrac{\sigma}{\epsilon} \log \dfrac{1}{\delta}, & k = 2, \\ \left(\dfrac{\sigma}{\epsilon}\right)^{k/(k-1)} \log \dfrac{1}{\delta}, & 1 < k < 2. \end{cases}$$

We now specify the fixed protocol whose horizon is bounded by this quantity.

Localization block. Before any response, split the sample indices into disjoint fixed sets I_{loc} and I_{ref} of cardinalities N_{loc} and N_{ref}. On I_{loc} use the coding-based fully non-adaptive localization construction of Lau and Scarlett at confidence $\eta = \delta/4$ (Lau and Scarlett, 2026, Theorem 16 and Appendix "Two-stage estimator"). Its public seed is R_{loc}, its precommitted query sets are $\mathcal{B}_i(R_{\mathrm{loc}})$, and its bits and always-defined decoder output are

$$Y_i^{\mathrm{loc}} = \mathbf{1}\{X_i \in \mathcal{B}_i(R_{\mathrm{loc}})\}, \qquad c = \mathsf{Dec}_{\mathrm{loc}}\big(R_{\mathrm{loc}}, (Y_i^{\mathrm{loc}})_{i \in I_{\mathrm{loc}}}\big).$$

Scales and dyadic geometry. For positive constants a_k, b_k depending only on k, put

$$h_0 = a_k \sigma, \qquad H_\star = b_k \sigma \left(\frac{\sigma}{\epsilon}\right)^{1/(k-1)}, \qquad S = \left\lceil \log_2 \frac{H_\star}{h_0} \right\rceil, \qquad h_s = 2^s h_0, \quad H = h_S.$$

The theorem chooses c_k so that $S \geq 1$. Define

$$Z_S = \sum_{s=0}^{S} h_s^{2-k}, \qquad p_s = \frac{h_s^{2-k}}{Z_S}, \quad 0 \leq s \leq S.$$

For $j \in \mathbb{Z}$, let

$$P_{s,j} = [j h_s, (j+1) h_s), \quad m_{s,j} = (j + \tfrac{1}{2}) h_s, \quad J_{s,j} = P_{s,j-1} \cup P_{s,j} \cup P_{s,j+1}.$$

For $s \geq 1$ and $b \in \{0, 1\}$ set

$$\mathcal{R}_{s,j,b} = J_{s,j} \setminus J_{s-1,2j+b}, \qquad \mathcal{R}_{0,j} = J_{0,j},$$

and write $\mathcal{J}_{s,\ell} = \{j \in \mathbb{Z} : j \equiv \ell \pmod 4\}$. The half-open convention fixes every boundary tie.

Precommitted refinement queries. For each $i \in I_{\mathrm{ref}}$ draw, before observing any message,

$$L_i \sim (p_0, \dots, p_S), \qquad C_i \sim \mathrm{Unif}\{0,1,2,3\}, \qquad U_i \sim \mathrm{Unif}[-1,1].$$

If $L_i = 0$, set $T_i = \mathrm{coord}$. If $L_i = s \ge 1$, also draw $T_i \sim \mathrm{Unif}\{\mathrm{coord}, \mathrm{mass}\}$ and $B_i \sim \mathrm{Unif}\{0,1\}$. Independently draw a countable Rademacher family $\rho_{i,s,j} \in \{-1,+1\}$ indexed by $0 \le s \le S$ and $j \in \mathbb{Z}$. At the selected level $s = L_i$, define

$$F_i(x) = \begin{cases} \displaystyle\sum_{j \in \mathcal{J}_{0,C_i}} \rho_{i,0,j} \frac{x - m_{0,j}}{2h_0} \mathbf{1}_{\mathcal{R}_{0,j}}(x), & L_i = 0, \\ \displaystyle\sum_{j \in \mathcal{J}_{s,C_i}} \rho_{i,s,j} \psi_{s,j,T_i}(x) \mathbf{1}_{\mathcal{R}_{s,j,B_i}}(x), & L_i = s \ge 1, \end{cases}$$

where $\psi_{s,j,\mathrm{coord}}(x) = (x - m_{s,j})/(2h_s)$ and $\psi_{s,j,\mathrm{mass}}(x) = 1$. The sole transmitted refinement bit and the decoder-computable baseline are

$$A_i = \{x : F_i(x) \ge U_i\}, \qquad Y_i = \mathbf{1}\{X_i \in A_i\}, \qquad Y_i^0 = \mathbf{1}\{0 \ge U_i\}, \qquad \Delta Y_i = Y_i - Y_i^0.$$

Decoder path and statistic. Only after both blocks have been collected, define

$$j_0(c) = \min \operatorname*{argmin}_{j \in \mathbb{Z}} |c - m_{0,j}|, \qquad m_0 = m_{0,j_0(c)},$$

and, for $s \ge 1$,

$$j_s = \left\lfloor \frac{j_0(c)}{2^s} \right\rfloor, \qquad b_s = j_{s-1} - 2j_s \in \{0,1\}, \qquad m_s = m_{s,j_s}, \qquad d_s = m_s - m_0.$$

Let $\kappa_s = j_s \bmod 4$ and

$$R_0(c) = \mathcal{R}_{0,j_0(c)}, \qquad R_s(c) = \mathcal{R}_{s,j_s,b_s}, \quad 1 \le s \le S.$$

For $i \in I_{\mathrm{ref}}$ define

$$W_i(c) = \begin{cases} \dfrac{16h_0}{p_0} \mathbf{1}\{C_i = \kappa_0\} \rho_{i,0,j_0(c)} \Delta Y_i, & L_i = 0, \\ \dfrac{16}{p_s} \mathbf{1}\{C_i = \kappa_s\} \mathbf{1}\{B_i = b_s\} \rho_{i,s,j_s} \\ \quad \times \left[4h_s \mathbf{1}\{T_i = \mathrm{coord}\} + 2d_s \mathbf{1}\{T_i = \mathrm{mass}\}\right] \Delta Y_i, & L_i = s \ge 1. \end{cases}$$

Fixed aggregation. For positive k-only constants α_k, β_k, set

$$G_\delta = 2\left\lceil \alpha_k \log \frac{8}{\delta} \right\rceil + 1, \qquad B_{\mathrm{ref}} = \left\lceil \beta_k \frac{\sigma^k Z_S}{\epsilon^2} \right\rceil, \qquad N_{\mathrm{ref}} = G_\delta B_{\mathrm{ref}}.$$

Partition I_{ref} in advance into G_δ consecutive groups of size B_{ref} and write

$$\overline{W}_g(c) = \frac{1}{B_{\mathrm{ref}}} \sum_{i \in G_g} W_i(c), \qquad \widehat{\mu} = m_0 + \operatorname*{median}_{1 \le g \le G_\delta} \overline{W}_g(c).$$

Thus the decoder-selected path changes only which already-transmitted bits are retained and how they are centered; it does not change a query.

3 Main Theorem

Theorem 3.1 (Order-optimal fully non-adaptive one-bit mean estimation)**.** *Under Assumptions 1–4, for every fixed $k > 1$ there are constants $c_k, C_k > 0$ and choices $a_k, b_k, \alpha_k, \beta_k > 0$, all depending only on k, such that the protocol of Section 2 is well defined for every known $\lambda \geq \sigma > 0$, every $0 < \epsilon \leq c_k\sigma$, and every $\delta \in (0, 1/2)$. Every query is a Borel set fixed before the first response; the protocol has no interactive transition, sends exactly one bit for each independent sample it uses, and has the deterministic non-stopping horizon $n = N_{\mathrm{loc}} + N_{\mathrm{ref}}$ satisfying*

$$n \leq C_k r_k(\lambda, \sigma, \epsilon, \delta).$$

Moreover, with probability taken jointly over all observations and all localization and refinement randomness,

$$\sup_{D \in \mathcal{D}(k,\lambda,\sigma)} \Pr_D \{|\widehat{\mu} - \mu(D)| > \epsilon\} \leq \delta.$$

The norm is absolute value on $\mathbb{R}$; the displayed rate retains every $\lambda, \sigma, \epsilon, \delta$ factor, and the hidden constant depends only on the fixed k.

For any realized localization output c, the same construction has two exact baseline reductions. If $D(J_{0,j_0(c)}) = 1$, all retained higher-level corrections and the outer residual vanish exactly, leaving the same fixed median aggregation applied only to the level-zero unbiased dither correction. If $D\{m_0(c)\} = 1$, every refinement correction vanishes seedwise and $\widehat{\mu} = m_0(c) = \mu(D)$.

Corollary 3.1 (Explicit three-regime rate specialization)**.** *Let C_k^{rec} and C_k^{var} be the finite k-only constants defined in Lemma A.2 and Proposition A.7. For fixed $k > 1$, choose*

$$a_k \geq 200, \quad b_k = \max\{a_k, (4C_k^{\mathrm{rec}})^{1/(k-1)}\}, \quad \beta_k = 16C_k^{\mathrm{var}}, \quad \alpha_k = 4,$$

and

$$0 < c_k \leq \min\left\{\frac{1}{2}, \left(\frac{b_k}{2a_k}\right)^{k-1}\right\}.$$

Then, for every known $\lambda \geq \sigma > 0$, every $0 < \epsilon \leq c_k\sigma$, and every $\delta \in (0, 1/2)$, the fully non-adaptive protocol precommits all queries, sends one bit per independent sample, and has deterministic non-stopping horizon $n = N_{\mathrm{loc}} + N_{\mathrm{ref}}$ satisfying

$$n \leq C_k \left[\log\frac{\lambda}{\sigma} + \begin{cases} \dfrac{\sigma^2}{\epsilon^2}\log\dfrac{1}{\delta}, & k > 2, \\ \dfrac{\sigma^2}{\epsilon^2}\log\dfrac{\sigma}{\epsilon}\log\dfrac{1}{\delta}, & k = 2, \\ \left(\dfrac{\sigma}{\epsilon}\right)^{k/(k-1)}\log\dfrac{1}{\delta}, & 1 < k < 2. \end{cases}\right]$$

Moreover,

$$\sup_{D \in \mathcal{D}(k,\lambda,\sigma)} \Pr_D \{|\widehat{\mu} - \mu(D)| > \epsilon\} \leq \delta.$$

The probability is unconditional and joint over all samples and protocol randomness, the norm is absolute value on $\mathbb{R}$, and C_k and every hidden constant depend only on the fixed k.

Proof. Theorem 3.1 gives the fixed-horizon protocol and its unconditional absolute-error guarantee. Proposition A.16 verifies the displayed auxiliary choices, all ceiling and probability conditions, and the stated three-regime bound with exactly the single $\log(\sigma/\epsilon)$ factor when $k = 2$. $\square$

4 Proof Sketch

The proof separates localization from refinement. Proposition A.1 instantiates the non-adaptive coding theorem of Lau and Scarlett (2026), maps its source hypotheses to our protocol, and recenters the kth moment around the decoded grid midpoint. The localization transcript is independent of the full refinement product array.

The dyadic construction is global and precommitted. Proposition A.3 shows that the decoder merely selects a nested chain from this fixed grid, whose child-subtracted rings partition the final padding. Lemma A.5 proves that every masked query is Borel and bounded for every realization of the countable masks.

Uniform dithering converts one bit into the value of a bounded query in expectation. Independent Rademacher masks cancel all same-color aliases, while the color, branch, type, and level weights invert their selection probabilities. Proposition A.5 therefore gives the exact truncated mean on the selected padding. Lemma A.10 charges every target and alias activation at x to $|x - m_0|^k$. Combining that charge with the exact dither square yields Proposition A.7; the target rings are disjoint, which is what prevents a second logarithm when $k = 2$.

Proposition A.9 chooses the terminal scale, controls the outer residual by the recentered moment, and evaluates the geometric sum Z_S in all three regimes. Conditional on the complete localization transcript, refinement observations remain independent and identically distributed. Chebyshev's inequality within each group and Hoeffding's inequality across groups then give Proposition A.11. Finally, Propositions A.12 and A.13 certify the communication model, Proposition A.14 integrates the conditional guarantee, and Proposition A.16 accounts for every ceiling and specializes the exact three-regime rate.

References

Lau, I. and Scarlett, J. (2026). Order-optimal sequential 1-bit mean estimation in general tail regimes. ArXiv:2604.07796v2. 2, 5, 7

A Proof Details

A.1 Coding localization, grid rounding, and recentering

Proposition A.1 (Verified non-adaptive localization and scalar wrapper)**.** *Under Assumptions 1–4, set $\eta = \delta/4$. There is a deterministic precommitted localization block using one Borel-set bit per sample and a decoder defined on every transcript such that, with $c = \mathrm{mid}(I)$ and $\mu = \mu(D)$,*

$$\mathcal{E}_{\mathrm{loc}} := \{|c - \mu| \le 50\sigma\}, \qquad \Pr_D(\mathcal{E}_{\mathrm{loc}}) \ge 1 - \frac{\delta}{4}.$$

Moreover, with $C_{\mathrm{loc},k} = 10001$,

$$N_{\mathrm{loc}} \le C_{\mathrm{loc},k}\left[1 + \log\frac{\lambda}{\sigma} + \log\frac{4}{\delta}\right].$$

In the trivial source branch $I = [-\lambda, \lambda]$, $c = 0$, and $N_{\mathrm{loc}} = 0$. In the nontrivial branch the fixed minimum-index Hamming tie rule, I, and c remain defined even when localization fails.

Proof. We instantiate Theorem 16 of Lau and Scarlett (2026), whose source label is `thm: alternative localization`, together with the construction in the source appendix labeled `appendix: two-stage`. That result assumes $\lambda \ge \sigma > 0$, $\eta \in (0, 1/2)$, iid observations with $\mu \in [-\lambda, \lambda]$, and $\mathbb{E}|X - \mu| \le \sigma$. Assumptions 1 and 2 give all but the last condition. Holder's inequality supplies it directly:

$$\mathbb{E}|X - \mu| \le \left(\mathbb{E}|X - \mu|^k\right)^{1/k} \left(\mathbb{E}1^{k/(k-1)}\right)^{(k-1)/k} \le \sigma.$$

Assumption 3 gives the iid source block, and Assumption 4 gives its fixed placement in the full protocol.

For completeness, the numerical source interface is as follows. Put $h_{\mathrm{src}} = 20\sigma$. If $2\lambda \le h_{\mathrm{src}}$, the construction asks no query and returns $I = [-\lambda, \lambda]$. Thus

$$|I| = 2\lambda \le 20\sigma, \qquad |c - \mu| = |\mu| \le \lambda \le 10\sigma,$$

so $\mathcal{E}_{\mathrm{loc}}$ occurs deterministically and $N_{\mathrm{loc}} = 0$.

If $2\lambda > h_{\mathrm{src}}$, define

$$N_{\mathrm{src}} = \left\lceil \frac{2\lambda}{h_{\mathrm{src}}} \right\rceil, \qquad \Delta = \frac{2\lambda}{N_{\mathrm{src}}}, \qquad \ell = \left\lceil 10000\left(\log N_{\mathrm{src}} + \log\frac{1}{\eta}\right)\right\rceil.$$

Then $h_{\mathrm{src}}/2 \le \Delta \le h_{\mathrm{src}}$. With $e_j^{\mathrm{src}} = -\lambda + (j-1)\Delta$, use the half-open bins $K_j^{\mathrm{src}} = [e_j^{\mathrm{src}}, e_{j+1}^{\mathrm{src}})$, closing only the final right endpoint, and clip observations outside $[-\lambda, \lambda]$ to the first or last bin. Hence every boundary has a unique bin index. The source fixes a deterministic balanced binary codeword of length ℓ for every bin, queries coordinate t of the codeword of the clipped bin containing X_t, decodes a minimum-Hamming-score bin with the minimum-index tie rule, and returns the union of that bin and at most its two neighbors on each side. The interval is therefore defined on every word and obeys

$$|I| \le 5\Delta \le 100\sigma.$$

The source guarantee gives $\Pr_D\{\mu \in I\} \ge 1 - \eta = 1 - \delta/4$. On $\{\mu \in I\}$,

$$|c - \mu| \le \frac{|I|}{2} \le 50\sigma,$$

which proves the required probability statement. The source's location-dependent refinement stage is not used.

To map each source bit into our query model, fix the promised codebook, take R_{loc} to be degenerate, and put $\mathcal{B}_t = Q_t^{-1}(\{1\})$. The clipped-bin map is constant on finitely many half-open or closed intervals and on the two outer rays; hence every $\mathcal{B}_t$ is Borel and $Y_t^{\mathrm{loc}} = \mathbf{1}\{X_t \in \mathcal{B}_t\} = Q_t(X_t)$. The bins, queries, codebook, sample count, and tie rule depend only on the known parameters and are fixed before any response.

Finally, in the nontrivial branch

$$N_{\mathrm{src}} = \left\lceil \frac{\lambda}{10\sigma} \right\rceil \geq 2.$$

For $x = \lambda/(10\sigma) > 1$,

$$N_{\mathrm{src}} = \lceil x \rceil \leq x + 1 \leq 2x = \frac{\lambda}{5\sigma} \leq \frac{\lambda}{\sigma},$$

and therefore

$$\begin{aligned} N_{\mathrm{loc}} &\leq 1 + 10000 \left(\log \frac{\lambda}{\sigma} + \log \frac{4}{\delta} \right) \\ &\leq 10001 \left(1 + \log \frac{\lambda}{\sigma} + \log \frac{4}{\delta} \right). \end{aligned}$$

This proves the proposition in both branches, including every tie and failure transcript. □

Lemma A.1 (Midpoint-to-grid core bridge)**.** *Under Assumption 1 and Proposition A.1, define $h_0 = a_k\sigma$ and use the minimum-index nearest-center rule. Every realized c satisfies*

$$|c - m_0| \leq \frac{h_0}{2}.$$

If also $|c - \mu(D)| \leq 50\sigma$ and $a_k \geq 200$, then

$$|m_0 - \mu(D)| \leq \frac{3h_0}{4}, \qquad \mu(D) \in [m_0 - 3h_0/4, m_0 + 3h_0/4] \subseteq J_{0,j_0(c)}.$$

The conclusion includes both source branches and every grid tie.

Proof. Let $q = \lfloor c/h_0 \rfloor$. The candidate $m_{0,q} = (q + 1/2)h_0$ is at distance at most $h_0/2$ from c, so the same is true of a nearest center. At a Voronoi boundary both candidates are at exactly that distance and the smaller-index rule preserves the bound. In fact

$$j_0(c) = \left\lceil \frac{c}{h_0} \right\rceil - 1,$$

so the rule is Borel and also covers negative c.

On $\mathcal{E}_{\mathrm{loc}}$, Proposition A.1 and $a_k \geq 200$ give

$$\begin{aligned} |m_0 - \mu(D)| &\leq |m_0 - c| + |c - \mu(D)| \\ &\leq \frac{h_0}{2} + 50\sigma = \left(\frac{1}{2} + \frac{50}{a_k} \right) h_0 \leq \frac{3h_0}{4}. \end{aligned}$$

The closed core is contained in $J_{0,j_0} = [m_0 - 3h_0/2, m_0 + 3h_0/2)$, including its two core endpoints. □

Lemma A.2 (Recentered moment certificate)**.** *Under Assumptions 1 and 2 and Lemma A.1, if* $|c - \mu(D)| \le 50\sigma$ *and* $a_k \ge 200$*, then*

$$M_k(c) := \int_{\mathbb{R}} |x - m_0(c)|^k D(dx) \le C_k^{\mathrm{rec}} \sigma^k, \qquad C_k^{\mathrm{rec}} := 2^{k-1} \left[1 + \left(\frac{3a_k}{4} \right)^k \right].$$

In particular, the recentered moment is finite and its constant depends only on the fixed k*.*

Proof. For fixed c on $\mathcal{E}_{\mathrm{loc}}$, the power-triangle inequality gives

$$|x - m_0|^k \le 2^{k-1}\big(|x - \mu(D)|^k + |\mu(D) - m_0|^k\big).$$

Indeed, this follows from convexity because $(|u|+|v|)^k \le 2^{k-1}(|u|^k+|v|^k)$. Integrating and applying Assumption 2 and Lemma A.1 yields

$$\begin{aligned} M_k(c) &\le 2^{k-1} \left(\mathbb{E}_D |X - \mu(D)|^k + |\mu(D) - m_0|^k \right) \\ &\le 2^{k-1} \left[\sigma^k + (3h_0/4)^k \right] = C_k^{\mathrm{rec}} \sigma^k. \end{aligned}$$

□

Proposition A.2 (Independent refinement product law)**.** *Under Assumptions 1– 4 and Proposition A.1, let* $\mathscr{L}_{\mathrm{loc}}$ *be the sigma-field generated by the complete localization block, including its local samples, seed, fixed queries, bits, interval* I*, and outputs* c, m_0*. It is independent of the sigma-field generated by all refinement samples, levels, colors, branches, types, masks, dithers, and group assignments. The variables* c, m_0*,* $\mathcal{E}_{\mathrm{loc}}$ *and, for fixed* D*,* $M_k(c)$ *are* $\mathscr{L}_{\mathrm{loc}}$*-measurable. Conditional on* $\mathscr{L}_{\mathrm{loc}}$*, the per-sample refinement tuples retain their original independent product law; for every bounded measurable function* g *of the complete refinement block,*

$$\mathbb{E}_D[g \mid \mathscr{L}_{\mathrm{loc}}] = \mathbb{E}_D g \quad \textit{almost surely.}$$

Proof. Every localization query, bit, I, c, and m_0 is a measurable function of the localization samples and the deterministic source objects. The event $\mathcal{E}_{\mathrm{loc}}$ is therefore $\mathscr{L}_{\mathrm{loc}}$-measurable. For fixed D, consider

$$\Phi_D(m) = \left(\int_{\mathbb{R}} |x - m|^k D(dx) \right)^{1/k}.$$

Minkowski's inequality and Assumption 2 give

$$\Phi_D(m) \le \|X - \mu(D)\|_{L_k(D)} + |m - \mu(D)| \le \sigma + |m - \mu(D)| < \infty.$$

The reverse triangle inequality in $L_k(D)$ gives, for all $m, m' \in \mathbb{R}$,

$$|\Phi_D(m) - \Phi_D(m')| \le \|(X - m) - (X - m')\|_{L_k(D)} = |m - m'|.$$

Thus Φ_D is Borel. Since $c \mapsto m_0(c)$ is Borel by Lemma A.1, $M_k(c) = \Phi_D(m_0(c))^k$ is $\mathscr{L}_{\mathrm{loc}}$-measurable.

Assumption 3 makes the complete refinement sample-and-seed family independent of the complete localization block. Measurable localization transformations preserve that independence. Conditioning on the first sigma-field therefore leaves the second sigma-field's joint law unchanged; this law is a product across sample indices by the same assumption and the precommitted construction. The conditional expectation identity follows, including on localization-failure transcripts.

Together, Proposition A.1, Lemmas A.1 and A.2, and this product law give the complete localization interface: midpoint radius 50σ, the selected-core certificate, the explicit recentered moment, and an independent refinement array, with the source and grid ceilings already included in N_{loc}. □

A.2 Dyadic ancestry, ring partition, and Borel query geometry

Lemma A.3 (Dyadic ancestor and outer-radius geometry)**.** *Under Assumption 1 and Proposition A.1, let $c \in \mathbb{R}$ be any realized decoder value and define j_0, j_s, b_s, m_s, d_s as in Section 2. For every $1 \le s \le S$,*

$$j_{s-1} = 2j_s + b_s, \qquad b_s \in \{0,1\}, \qquad J_{s-1,j_{s-1}} \subseteq J_{s,j_s}.$$

Moreover,

$$|d_s| < \frac{h_s}{2} \le h_s, \qquad [m_0 - H, m_0 + H] \subset J_{S,j_S}.$$

These statements hold for positive and negative indices, both child values, and every nearest-center or half-open-grid tie.

Proof. Put $t = c/h_0$. The integer $j_0 = \lceil t \rceil - 1$ satisfies $j_0 < t \le j_0 + 1$. If the last inequality is strict, its center is the unique nearest center; if equality holds, the centers indexed by j_0 and $j_0 + 1$ tie and j_0 is selected. This covers negative t as well.

For $s \ge 1$ define the Euclidean remainder

$$r_s = j_0 - 2^s \left\lfloor \frac{j_0}{2^s} \right\rfloor = j_0 - 2^s j_s.$$

The floor inequalities give $r_s \in \{0, 1, \ldots, 2^s - 1\}$, including for $j_0 < 0$, and hence

$$\begin{aligned} j_{s-1} &= \left\lfloor \frac{2^s j_s + r_s}{2^{s-1}} \right\rfloor = 2j_s + \left\lfloor \frac{r_s}{2^{s-1}} \right\rfloor, \\ b_s &= \left\lfloor \frac{r_s}{2^{s-1}} \right\rfloor \in \{0,1\}. \end{aligned}$$

Since $h_s = 2h_{s-1}$, direct endpoint calculation gives

$$\begin{aligned} J_{s,j} &= [(j-1)h_s, (j+2)h_s), \\ J_{s-1,2j} &= [(j-\tfrac{1}{2})h_s, (j+1)h_s), \\ J_{s-1,2j+1} &= [jh_s, (j+\tfrac{3}{2})h_s). \end{aligned}$$

Both child paddings lie in $J_{s,j}$. Substitution of j_s, b_s and iteration prove $J_{0,j_0} \subseteq \cdots \subseteq J_{S,j_S}$.

The same remainder identity yields

$$m_0 = (j_s + \vartheta_s)h_s, \qquad \vartheta_s = \frac{r_s + 1/2}{2^s} \in (0,1).$$

Consequently

$$d_s = (\tfrac{1}{2} - \vartheta_s)h_s, \qquad |d_s| < h_s/2.$$

At $s = S$, $H = h_S$ and

$$(j_S - 1)H < m_0 - H \le m_0 + H < (j_S + 2)H.$$

Thus both endpoints of the closed radius-H interval lie in the half-open padding J_{S,j_S}. $\square$

Proposition A.3 (Exact padded-ring partition)**.** *Under Assumption 1 and Lemma A.3, for every $c \in \mathbb{R}$ the target rings are pairwise disjoint and*

$$\dot{\bigcup_{s=0}^{S}} R_s(c) = J_{S,j_S}, \qquad \sum_{s=0}^{S} \mathbf{1}\{x \in R_s(c)\} = \mathbf{1}\{x \in J_{S,j_S}\}$$

for every $x \in \mathbb{R}$. Every $x \in J_{0,j_0}$ belongs to no target ring $R_s(c)$ with $s \ge 1$.

Proof. Write $\mathbf{J}_s = J_{s,j_s}$. The exact child identity in Lemma A.3 gives, for $s \geq 1$,

$$R_s(c) = \mathcal{R}_{s,j_s,b_s} = J_{s,j_s} \setminus J_{s-1,2j_s+b_s} = \mathbf{J}_s \setminus \mathbf{J}_{s-1},$$

whereas $R_0(c) = \mathbf{J}_0$. Therefore

$$\mathbf{J}_s = \mathbf{J}_{s-1} \,\dot\cup\, R_s(c).$$

Iterating this identity yields the asserted disjoint union. Equivalently, if $t < s$, then $R_t(c) \subseteq \mathbf{J}_t \subseteq \mathbf{J}_{s-1}$, whereas $R_s(c) \cap \mathbf{J}_{s-1} = \varnothing$. This also proves the pointwise indicator identity at every half-open endpoint. Finally, if $x \in \mathbf{J}_0$, nesting puts x in every $\mathbf{J}_{s-1}$, so it is in none of the differences $\mathbf{J}_s \setminus \mathbf{J}_{s-1}$. □

Lemma A.4 (Four-color support, endpoint, and amplitude geometry)**.** *Under Assumption 1 and Lemma A.3, fix $0 \leq s \leq S$ and $\ell \in \{0, 1, 2, 3\}$. The paddings $\{J_{s,j} : j \equiv \ell \pmod 4\}$ are pairwise disjoint, and so are the rings at any fixed branch. All these sets are Borel and*

$$\left|\frac{x - m_{s,j}}{2h_s}\right| \leq \frac{3}{4} \quad \textit{on every such ring,}$$

while the mass amplitude is one. If $s \geq 1$ and $x \in J_{0,j_0}$, then x lies in neither the target ring nor any alias $\mathcal{R}_{s,j,b_s}$ with $j \equiv j_s \pmod 4$.

Proof. The padding is $J_{s,j} = [(j-1)h_s, (j+2)h_s)$. If $j < j'$ have the same residue modulo four, then $j' \geq j + 4$, so the right endpoint of the first padding is $(j+2)h_s$ and the left endpoint of the second is at least $(j+3)h_s$. They are disjoint with a gap of at least one cell. This calculation is unchanged for negative indices.

For $s \geq 1$, the child-padding formulas give

$$\begin{aligned}
\mathcal{R}_{s,j,0} &= [(j-1)h_s, (j-\tfrac{1}{2})h_s) \cup [(j+1)h_s, (j+2)h_s), \\
\mathcal{R}_{s,j,1} &= [(j-1)h_s, jh_s) \cup [(j+\tfrac{3}{2})h_s, (j+2)h_s).
\end{aligned}$$

At level zero, $\mathcal{R}_{0,j} = J_{0,j}$. Thus all sets are Borel and the formulas settle each boundary. For $x \in J_{s,j}$,

$$-\frac{3h_s}{2} \leq x - m_{s,j} < \frac{3h_s}{2},$$

which proves the coordinate bound after division by $2h_s$; equality at the included left endpoint is allowed.

For the final assertion, take $x \in J_{0,j_0}$. Nesting puts it in J_{s,j_s} and, more strongly, in $J_{s-1,j_{s-1}}$, excluding it from $\mathcal{R}_{s,j_s,b_s}$. Every other $j \equiv j_s \pmod 4$ has padding disjoint from J_{s,j_s}, so its ring cannot contain x. These cases exhaust the retained color and branch. □

Lemma A.5 (Jointly Borel bounded precommitted query map)**.** *Under Assumptions 1 and 4 and Lemma A.4, the formula for each F_i is jointly Borel in the sample and its complete precommitted level, color, type, branch, and countable mask seed. For every seed realization and every $x \in \mathbb{R}$,*

$$|F_i(x)| \leq 1.$$

Consequently each realized F_i is Borel and every dithered superlevel set $\{x : F_i(x) \geq u\}$, $u \in [-1, 1]$, is Borel. Neither map contains the decoder output c.

Proof. For fixed i, let

$$\mathcal{M}_i = \{-1,+1\}^{\{0,\dots,S\}\times\mathbb{Z}}$$

carry its countable product Borel sigma-field. The query-seed space is the finite disjoint union

$$\Omega_i^{\mathrm{qry}} := \big(\{0\}\times\{0,1,2,3\}\times\mathcal{M}_i\big) \\ \dot\cup \dot{\bigcup_{s=1}^{S}}\big(\{s\}\times\{0,1,2,3\}\times\{\mathrm{coord},\mathrm{mass}\}\times\{0,1\}\times\mathcal{M}_i\big),$$

with finite coordinates discrete. The level-zero component correctly has no child variable.

Fix s,ℓ and truncate the relevant sum to indices $j\equiv\ell \pmod 4$ with $|j|\le N$. Each summand is the product of a mask coordinate, a continuous affine function or the constant one, and the indicator of a Borel ring from Lemma A.4. Every finite partial sum is jointly Borel. Same-color disjointness implies that at each (x,ρ) at most one summand of the full countable sum is nonzero; the partial sums therefore stabilize pointwise. Their limit is jointly Borel. Finite case distinctions over L_i, C_i, T_i, B_i preserve this property.

The same uniqueness proves the range bound. With no active ring the value is zero; on a coordinate ring the unique summand has magnitude at most $3/4$; on a mass ring it is a Rademacher sign and has magnitude one. This covers every mask realization and half-open endpoint without an absolute-summability assumption.

For fixed seed ω, $x\mapsto F_i(x,\omega)$ is Borel, and more strongly

$$\{(x,\omega,u) : F_i(x,\omega)-u\ge 0\}$$

is Borel in $\mathbb{R}\times\Omega_i^{\mathrm{qry}}\times[-1,1]$. Assumption 4 places ω,u before the first response, and the formula contains no c or response bit.

Lemma A.3, Proposition A.3, Lemma A.4, and this measurability result jointly give the negative-index-safe nested path, exact ring partition, retained-ring inactivity on J_{0,j_0}, and a bounded Borel query for every countable-mask realization. □

A.3 Dither inversion, mask projection, and the mean telescope

Lemma A.6 (Exact centered-uniform dither moments)**.** *Under Assumption 3, let $U\sim\mathrm{Unif}[-1,1]$ and fix $f\in[-1,1]$ independently of U. Define*

$$\Delta_f(U) = \mathbf{1}\{f\ge U\} - \mathbf{1}\{0\ge U\}.$$

Then, including $f=-1,0,1$,

$$\mathbb{E}_U\Delta_f(U) = \frac{f}{2}, \qquad \mathbb{E}_U\Delta_f(U)^2 = \frac{|f|}{2}.$$

If a random $F\in[-1,1]$ is independent of U, both identities hold conditionally on F.

Proof. For $0\le f\le 1$, $\Delta_f(u) = \mathbf{1}_{(0,f]}(u)$, whereas for $-1\le f<0$, $\Delta_f(u) = -\mathbf{1}_{(f,0]}(u)$. At $f=0$ the two indicators agree identically. Uniform measure on $[-1,1]$ has density $1/2$, so integrating the signed interval gives $f/2$, while squaring removes its sign and gives $|f|/2$. The threshold endpoints are correct and have zero uniform mass. Conditioning on an independent random F fixes its value and applies the same pointwise calculation. □

Lemma A.7 (Fresh-mask projection and exact alias cancellation)**.** *Under Assumption 3, Lemmas A.6, A.4, and A.5, fix a level, color class, and, at a higher level, branch and type. Let $(E_j)_{j\in\mathcal{J}}$ be the resulting pairwise-disjoint rings, let $(\rho_j)_{j\in\mathcal{J}}$ be independent Rademacher masks, and write*

$$F_\rho(x) = \sum_{j\in\mathcal{J}} \rho_j a_j(x)\mathbf{1}_{E_j}(x) \in [-1, 1].$$

For any target $j_\star \in \mathcal{J}$ and independent uniform dither U,

$$\mathbb{E}_{\rho,U}\left[\rho_{j_\star}\Delta_{F_\rho(x)}(U)\right] = \frac{1}{2}a_{j_\star}(x)\mathbf{1}_{E_{j_\star}}(x)$$

for every x. In particular, a uniquely active alias $j \neq j_\star$ contributes exactly zero.

Proof. Conditioning first on the complete mask family and applying Lemma A.6 gives

$$\mathbb{E}_{\rho,U}[\rho_{j_\star}\Delta_{F_\rho(x)}(U)] = \frac{1}{2}\mathbb{E}_\rho[\rho_{j_\star}F_\rho(x)].$$

At fixed x, Lemma A.4 leaves either no active summand or one active index q. In the first case both sides vanish. In the second, the last expectation is

$$\frac{1}{2}a_q(x)\mathbf{1}_{E_q}(x)\mathbb{E}[\rho_{j_\star}\rho_q] = \frac{1}{2}a_q(x)\mathbf{1}_{E_q}(x)\mathbf{1}\{q = j_\star\},$$

because a mask squares to one and distinct masks are independent and centered. No countable sum is exchanged with expectation. □

Proposition A.4 (Exact per-level importance inversion)**.** *Under Assumptions 3 and 4, Propositions A.2 and A.3, and Lemmas A.3, A.4, A.5, and A.7, fix a decoder value c independent of the refinement block. For every $x \in \mathbb{R}$, put*

$$K_s(x, c) = \mathbb{E}[W_i(c)\mathbf{1}\{L_i = s\} \mid X_i = x, c].$$

Then

$$K_0(x, c) = (x - m_0)\mathbf{1}_{R_0(c)}(x),$$

and, for $1 \le s \le S$,

$$K_s(x, c) = \left[(x - m_s) + d_s\right]\mathbf{1}_{R_s(c)}(x) = (x - m_0)\mathbf{1}_{R_s(c)}(x).$$

Every same-color alias contributes zero.

Proof. Fix c, x. Proposition A.2 leaves every level and seed law unchanged and

$$\Delta Y_i = \mathbf{1}\{F_i(x) \ge U_i\} - \mathbf{1}\{0 \ge U_i\} = \Delta_{F_i(x)}(U_i).$$

At level zero, $W_i(c)$ vanishes unless $C_i = \kappa_0$, an event of probability $1/4$. Conditional on it, Lemma A.7, with target j_0 and amplitude $(x - m_{0,j})/(2h_0)$, yields

$$\mathbb{E}_{\rho,U}[\rho_{i,0,j_0}\Delta Y_i \mid x, c, L_i = 0, C_i = \kappa_0] = \frac{x - m_0}{4h_0}\mathbf{1}_{R_0(c)}(x).$$

All aliases have already vanished. Substitution gives the complete ledger

$$\begin{aligned} K_0(x, c) &= p_0 \cdot \frac{1}{4} \cdot \frac{16h_0}{p_0} \cdot \frac{x - m_0}{4h_0}\mathbf{1}_{R_0(c)}(x) \\ &= (x - m_0)\mathbf{1}_{R_0(c)}(x). \end{aligned}$$

Thus the coefficient inverts the level, color, dither, and coordinate normalization; no branch or type exists at level zero.

Now fix $s \geq 1$. The statistic vanishes unless $C_i = \kappa_s$ and $B_i = b_s$, whose probabilities are $1/4$ and $1/2$. For coordinate type, Lemma A.7 gives

$$\mathbb{E}_{\rho,U}[\rho_{i,s,j_s}\Delta Y_i \mid x, c, L_i = s, C_i = \kappa_s, B_i = b_s, T_i = \text{coord}] = \frac{x - m_s}{4h_s}\mathbf{1}_{R_s(c)}(x).$$

For mass type it gives

$$\mathbb{E}_{\rho,U}[\rho_{i,s,j_s}\Delta Y_i \mid x, c, L_i = s, C_i = \kappa_s, B_i = b_s, T_i = \text{mass}] = \frac{1}{2}\mathbf{1}_{R_s(c)}(x).$$

Both expressions vanish for an active alias. Since each type has probability $1/2$, the coordinate contribution is

$$\begin{aligned} p_s \cdot \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{16}{p_s} \cdot 4h_s \cdot \frac{x - m_s}{4h_s}\mathbf{1}_{R_s(c)}(x) \\ = (x - m_s)\mathbf{1}_{R_s(c)}(x), \end{aligned}$$

and the mass contribution is

$$\begin{aligned} p_s \cdot \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{16}{p_s} \cdot 2d_s \cdot \frac{1}{2}\mathbf{1}_{R_s(c)}(x) \\ = d_s\mathbf{1}_{R_s(c)}(x). \end{aligned}$$

These displays invert every level, color, branch, type, dither, and coordinate factor separately. Adding the mutually exclusive types and using $d_s = m_s - m_0$ proves the result, including $d_s = 0$. □

Proposition A.5 (Exact padded-ring mean telescope and residual)**.** *Under Assumptions 3 and 4, Propositions A.2, A.3, and A.4, and Lemma A.4, every decoder value c independent of the refinement block satisfies, for all x,*

$$\mathbb{E}[W_i(c) \mid X_i = x, c] = (x - m_0)\mathbf{1}\{x \in J_{S,j_S}\}.$$

Writing $\theta(c) = \mathbb{E}[W_i(c) \mid c]$, one has

$$\theta(c) = \mathbb{E}_D[(X - m_0)\mathbf{1}\{X \in J_{S,j_S}\} \mid c]$$

and

$$\mu(D) - m_0 = \theta(c) + \mathbb{E}_D[(X - m_0)\mathbf{1}\{X \notin J_{S,j_S}\} \mid c].$$

There is no localization, fine-scale, ring-gap, or alias residual. If $D(J_{0,j_0}) = 1$, every retained higher correction is seedwise zero and the outer residual is zero. If $D\{m_0\} = 1$, then $W_i(c) = 0$ for every seed realization.

Proof. The level events form a finite partition. Proposition A.4 and the exact ring partition give pointwise

$$\begin{aligned} \mathbb{E}[W_i(c) \mid X_i = x, c] &= \sum_{s=0}^{S} K_s(x, c) \\ &= (x - m_0)\sum_{s=0}^{S}\mathbf{1}_{R_s(c)}(x) \\ &= (x - m_0)\mathbf{1}_{J_{S,j_S}}(x). \end{aligned}$$

This equality is exact at every half-open boundary. A same-color alias is zero already by Lemma A.7.

Proposition A.2 and the tower rule now yield

$$\theta(c) = \int_{J_{S,j_S}} (x - m_0) D(dx).$$

For fixed c, m_0, J_{S,j_S} are deterministic, and $X - m_0$ is integrable. Splitting its integral gives

$$\begin{aligned}\mu(D) - m_0 &= \int_{\mathbb{R}} (x - m_0) D(dx) \\ &= \int_{J_{S,j_S}} (x - m_0) D(dx) + \int_{J^c_{S,j_S}} (x - m_0) D(dx),\end{aligned}$$

so the second integral is the sole residual.

If $D(J_{0,j_0}) = 1$, Lemma A.4 shows that for every $s \geq 1$ neither the target nor any retained alias ring can be active. A mismatched retained color or branch makes $W_i(c) = 0$ directly; when they match, $F_i(X_i) = 0$ and hence $\Delta Y_i = 0$. Thus every higher correction is seedwise zero. Nesting gives $J_{0,j_0} \subseteq J_{S,j_S}$, so the outer residual is also zero and the level-zero identity supplies $\theta(c) = \mu(D) - m_0$. If $X_i = m_0$, the target level-zero coordinate is zero as well, so every retained ΔY_i and every $W_i(c)$ vanishes for all seeds.

Thus Lemmas A.6 and A.7, Proposition A.4, and the preceding finite telescope account for every dither, mask, level, color, branch, and type factor and leave exactly the displayed outer residual, with both exact baselines intact. □

A.4 Target and alias distance with the all-ring activation ledger

Lemma A.8 (Target-child exclusion forces residual distance)**.** *Under Assumption 1 and Lemma A.3, let $c, x \in \mathbb{R}$ and $1 \leq s \leq S$. If*

$$x \in \mathcal{R}_{s,j_s,b_s} = J_{s,j_s} \setminus J_{s-1,j_{s-1}},$$

then

$$|x - m_0| > \frac{h_s}{2}, \qquad h_s < 2|x - m_0|.$$

This holds for both branches, signs of the path index, and all half-open child-padding endpoints.

Proof. Define

$$a_s = j_0 - 2^s j_s = j_0 - 2^s \left\lfloor \frac{j_0}{2^s} \right\rfloor \in \{0, \ldots, 2^s - 1\}.$$

Then

$$m_0 = (j_s + \theta_s) h_s, \qquad \theta_s = \frac{a_s + 1/2}{2^s},$$

and

$$b_s = j_{s-1} - 2j_s = \left\lfloor \frac{a_s}{2^{s-1}} \right\rfloor.$$

These Euclidean-remainder identities remain valid for negative j_0.

If $b_s = 0$, then $0 \leq a_s \leq 2^{s-1} - 1$ and

$$\frac{1}{2^{s+1}} \leq \theta_s \leq \frac{1}{2} - \frac{1}{2^{s+1}}.$$

Since $J_{s-1,j_{s-1}} = [(j_s - \frac{1}{2})h_s, (j_s+1)h_s)$,

$$m_0 - \frac{h_s}{2} = (j_s - \tfrac{1}{2} + \theta_s)h_s > (j_s - \tfrac{1}{2})h_s,$$
$$m_0 + \frac{h_s}{2} = (j_s + \tfrac{1}{2} + \theta_s)h_s < (j_s + 1)h_s.$$

Thus the closed radius-$h_s/2$ interval lies strictly inside the selected child padding.

If $b_s = 1$, then $2^{s-1} \le a_s \le 2^s - 1$ and

$$\frac{1}{2} + \frac{1}{2^{s+1}} \le \theta_s \le 1 - \frac{1}{2^{s+1}}.$$

Now $J_{s-1,j_{s-1}} = [j_s h_s, (j_s + \frac{3}{2})h_s)$ and

$$m_0 - \frac{h_s}{2} = (j_s - \tfrac{1}{2} + \theta_s)h_s > j_s h_s,$$
$$m_0 + \frac{h_s}{2} = (j_s + \tfrac{1}{2} + \theta_s)h_s < (j_s + \tfrac{3}{2})h_s.$$

The same strict containment holds. In either branch, a point in the target ring lies outside the selected child padding and therefore outside this closed interval. Hence $|x - m_0| > h_s/2$. The strict containment also handles the asymmetric included-left and excluded-right endpoints. □

Lemma A.9 (Same-color aliases are separated from the path center)**.** *Under Assumption 1 and Lemmas A.3 and A.4, let $c, x \in \mathbb{R}$, $1 \le s \le S$, and $q \in \mathbb{Z} \setminus \{0\}$. If $x \in \mathcal{R}_{s,j_s+4q,b_s}$, then*

$$|x - m_0| > 2h_s, \qquad h_s < \frac{|x - m_0|}{2}.$$

The conclusion is uniform over the retained branch, all index signs, and all half-open boundaries.

Proof. Lemma A.3 places m_0 strictly in the path cell:

$$j_s h_s < m_0 < (j_s + 1)h_s.$$

Every branch ring lies in its padding, so

$$\mathcal{R}_{s,j_s+4q,b_s} \subseteq [(j_s + 4q - 1)h_s, (j_s + 4q + 2)h_s).$$

If $q \ge 1$, activation implies $x \ge (j_s + 4q - 1)h_s \ge (j_s + 3)h_s$; since $m_0 < (j_s + 1)h_s$, this gives $x - m_0 > 2h_s$. Strictness remains at the included left endpoint when $q = 1$.

If $q \le -1$, activation and the excluded right endpoint give $x < (j_s + 4q + 2)h_s \le (j_s - 2)h_s$. Since $m_0 > j_s h_s$, $m_0 - x > 2h_s$. This is strict for $q = -1$ because the alias padding is right-open and m_0 is strictly inside the path cell. The two cases exhaust all nonzero integers and are insensitive to the sign of j_s or the branch. □

Lemma A.10 (Uniform all-ring activation ledger)**.** *Under Assumptions 1 and 4, Lemmas A.3, A.4, A.8, and A.9, define*

$$\Gamma_s(c) = \bigcup_{q \in \mathbb{Z}} \mathcal{R}_{s,j_s+4q,b_s}, \qquad 1 \le s \le S.$$

For every c, x, with $r = |x - m_0|$,

$$x \in \Gamma_s(c) \implies h_s \le 2r,$$

and

$$\sum_{s=1}^{S} h_s^k \mathbf{1}\{x \in \Gamma_s(c)\} \le C_{\mathrm{act},k} r^k, \qquad C_{\mathrm{act},k} = \frac{2^k}{1-2^{-k}}.$$

At each level at most one ring is active, so the sum charges every retained target and alias. It is exactly zero when $r = 0$ and for every $x \in J_{0,j_0}$.

Proof. Every integer congruent to j_s modulo four is uniquely $j_s + 4q$. Lemma A.4 makes the active index unique at each level. For $q = 0$, Lemma A.8 gives $h_s < 2r$; for $q \neq 0$, Lemma A.9 gives the stronger $h_s < r/2$. Thus no retained ring is omitted and

$$\sum_{s=1}^{S} h_s^k \mathbf{1}\{x \in \Gamma_s(c)\} \le \sum_{\substack{1 \le s \le S \\ h_s \le 2r}} h_s^k.$$

If the index set is empty, the result is immediate. Otherwise let t be its largest element. Since $h_s = 2^{s-t} h_t$,

$$\begin{aligned}\sum_{\substack{1 \le s \le S \\ h_s \le 2r}} h_s^k &= \sum_{s=1}^{t} h_s^k = h_t^k \sum_{u=0}^{t-1} 2^{-ku} \\ &\le \frac{h_t^k}{1-2^{-k}} \le \frac{(2r)^k}{1-2^{-k}} = C_{\mathrm{act},k} r^k.\end{aligned}$$

The denominator is positive because $k > 1$, and the constant is independent of the path, branch, level count, decoder value, and sample point.

If $r = 0$, no positive h_s can satisfy $h_s \le 2r$, so the ledger is zero. If $x \in J_{0,j_0}$, nesting puts it in every selected child padding, so the target ring is inactive; it also lies in J_{s,j_s}, which is disjoint from every $q \neq 0$ alias padding. Hence the entire higher-level ledger vanishes pointwise, including at all half-open endpoints.

The target and alias distance lemmas exhaust the retained ring indices, and the geometric summation above therefore charges all higher-level target and alias activations with the explicit constant $C_{\mathrm{act},k}$. □

A.5 All-alias square ledgers and conditional variance

For this subsection only, write

$$E_{0,q}(c) = \mathcal{R}_{0,j_0+4q}, \qquad \Gamma_0(c) = \bigcup_{q \in \mathbb{Z}} E_{0,q}(c),$$

and, for $1 \le s \le S$,

$$E_{s,q}(c) = \mathcal{R}_{s,j_s+4q,b_s}, \qquad \Gamma_s(c) = \bigcup_{q \in \mathbb{Z}} E_{s,q}(c).$$

At fixed (s, x, c), Lemma A.4 makes at most one term over q nonzero.

Lemma A.11 (Exact level-zero all-alias square ledger)**.** *Under Assumptions 3 and 4, Proposition A.2, and Lemmas A.4, A.5, and A.6, every $c, x \in \mathbb{R}$ satisfies*

$$\begin{aligned}&\mathbb{E}[W_i(c)^2 \mathbf{1}\{L_i = 0\} \mid X_i = x, c] \\ &\quad = \frac{16 h_0}{p_0} \sum_{q \in \mathbb{Z}} |x - m_{0,j_0+4q}| \mathbf{1}_{E_{0,q}(c)}(x) \\ &\quad \le \frac{24 h_0^2}{p_0} \mathbf{1}_{\Gamma_0(c)}(x).\end{aligned}$$

The equality includes the target $q = 0$ and every level-zero alias.

Proof. On $\{L_i = 0, C_i = \kappa_0\}$, unique activity gives

$$F_i(x) = \sum_{q\in\mathbb{Z}} \rho_{i,0,j_0+4q} \frac{x - m_{0,j_0+4q}}{2h_0} \mathbf{1}_{E_{0,q}(c)}(x).$$

Although $W_i(c)$ uses the target mask, its square contains $\rho_{i,0,j_0}^2 = 1$. Conditional on all masks, the dither-square identity and unique activity yield

$$\mathbb{E}_{U_i}[\Delta Y_i^2] = \sum_{q\in\mathbb{Z}} \frac{|x - m_{0,j_0+4q}|}{4h_0} \mathbf{1}_{E_{0,q}(c)}(x).$$

No mask cancellation occurs in this nonnegative square; an alias produces the same absolute-amplitude term as the target.

Proposition A.2 retains $\Pr\{L_i = 0\} = p_0$ and $\Pr\{C_i = \kappa_0\} = 1/4$. Therefore

$$\begin{aligned} p_0 \cdot \frac{1}{4} \left(\frac{16h_0}{p_0}\right)^2 \sum_q \frac{|x - m_{0,j_0+4q}|}{4h_0} \mathbf{1}_{E_{0,q}(c)}(x) \\ = \frac{16h_0}{p_0} \sum_q |x - m_{0,j_0+4q}| \mathbf{1}_{E_{0,q}(c)}(x). \end{aligned}$$

The factor p_0^{-1} remains explicit. On an active level-zero ring, $|x - m_{0,j}| \le 3h_0/2$ by Lemma A.4; at most one q is active. This proves the factor-24 bound. No branch or type probability appears because neither variable exists at level zero. □

Lemma A.12 (Exact higher-level coordinate-mass all-alias square ledger)**.** *Under Assumptions 3 and 4, Proposition A.2, and Lemmas A.3, A.4, A.5, and A.6, every $c, x \in \mathbb{R}$ and $1 \le s \le S$ satisfy*

$$\begin{aligned} &\mathbb{E}[W_i(c)^2 \mathbf{1}\{L_i = s\} \mid X_i = x, c] \\ &= \frac{64h_s}{p_s} \sum_{q\in\mathbb{Z}} |x - m_{s,j_s+4q}| \mathbf{1}_{E_{s,q}(c)}(x) + \frac{32d_s^2}{p_s} \mathbf{1}_{\Gamma_s(c)}(x) \\ &\le \frac{128h_s^2}{p_s} \mathbf{1}_{\Gamma_s(c)}(x). \end{aligned}$$

Every target and alias appears in the exact sum; zero alias mean does not remove its square.

Proof. On the retained color and branch, the two types are mutually exclusive, so

$$[4h_s \mathbf{1}\{T_i = \text{coord}\} + 2d_s \mathbf{1}\{T_i = \text{mass}\}]^2 = 16h_s^2 \mathbf{1}\{T_i = \text{coord}\} + 4d_s^2 \mathbf{1}\{T_i = \text{mass}\}.$$

There is no coordinate-mass cross term. For coordinate type, unique activity and Lemma A.6 give

$$\mathbb{E}_{U_i}[\Delta Y_i^2] = \sum_q \frac{|x - m_{s,j_s+4q}|}{4h_s} \mathbf{1}_{E_{s,q}(c)}(x),$$

whereas for mass type they give

$$\mathbb{E}_{U_i}[\Delta Y_i^2] = \frac{1}{2} \mathbf{1}_{\Gamma_s(c)}(x).$$

These quantities are independent of mask signs, and the target mask in $W_i(c)^2$ squares to one even when an alias is active.

The exact probabilities of the level, retained color, retained branch, and either type are $p_s, 1/4, 1/2, 1/2$. Hence the coordinate square is

$$\begin{aligned} &p_s \cdot \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{1}{2} \left(\frac{16}{p_s} 4h_s\right)^2 \\ &\quad \times \sum_q \frac{|x - m_{s,j_s+4q}|}{4h_s} \mathbf{1}_{E_{s,q}(c)}(x) \\ &= \frac{64h_s}{p_s} \sum_q |x - m_{s,j_s+4q}| \\ &\quad \times \mathbf{1}_{E_{s,q}(c)}(x), \end{aligned}$$

and the mass square is

$$\begin{aligned} &p_s \cdot \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{1}{2} \left(\frac{16}{p_s} 2d_s\right)^2 \frac{1}{2} \mathbf{1}_{\Gamma_s(c)}(x) \\ &\quad = \frac{32d_s^2}{p_s} \mathbf{1}_{\Gamma_s(c)}(x). \end{aligned}$$

Exactly one factor p_s^{-1} remains after selection averaging.

On an active coordinate ring, $|x - m_{s,j}| \le 3h_s/2$, so the coordinate term is at most $96h_s^2 p_s^{-1} \mathbf{1}_{\Gamma_s(c)}(x)$. The displacement bound $|d_s| \le h_s$ makes the mass term at most $32h_s^2 p_s^{-1} \mathbf{1}_{\Gamma_s(c)}(x)$. Their sum proves the stated constant. The exact mass term vanishes when $d_s = 0$. □

Proposition A.6 (Pointwise all-level second-moment budget)**.** *Under Assumptions 3 and 4, Lemmas A.11, A.12, and A.10, every fixed $c, x \in \mathbb{R}$ satisfies*

$$\mathbb{E}[W_i(c)^2 \mid X_i = x, c] \le Z_S \left[24h_0^k + 128C_{\text{act},k}|x - m_0|^k\right].$$

This inequality sums all target and alias squares before taking expectation over X_i.

Proof. The level events are mutually exclusive and exhaustive. Lemmas A.11 and A.12 give

$$\mathbb{E}[W_i(c)^2 \mid X_i = x, c] \le \frac{24h_0^2}{p_0} \mathbf{1}_{\Gamma_0(c)}(x) + 128 \sum_{s=1}^{S} \frac{h_s^2}{p_s} \mathbf{1}_{\Gamma_s(c)}(x).$$

For every level,

$$p_s^{-1} = Z_S h_s^{k-2}, \qquad \frac{h_s^2}{p_s} = Z_S h_s^k.$$

Substitution without asymptotic notation yields

$$\begin{aligned} \mathbb{E}[W_i(c)^2 \mid X_i = x, c] &\le Z_S \left[24h_0^k \mathbf{1}_{\Gamma_0(c)}(x) + 128 \sum_{s=1}^{S} h_s^k \mathbf{1}_{\Gamma_s(c)}(x)\right] \\ &\le Z_S \left[24h_0^k + 128C_{\text{act},k}|x - m_0|^k\right]. \end{aligned}$$

The last line uses only $\mathbf{1}_{\Gamma_0(c)} \le 1$ and Lemma A.10. Every within-level ring sum has at most one active term and the level sum is finite, so no expectation or supremum is interchanged with a countable ring sum. □

Proposition A.7 (Conditional variance and the single middle-regime factor)**.** *Under Assumption 2, Propositions A.2 and A.6, and Lemma A.2, suppose $a_k \geq 200$ and fix a realized localization output with $|c - \mu(D)| \leq 50\sigma$. Then*
$$\mathrm{Var}(W_i(c) \mid c) \leq C_k^{\mathrm{var}} \sigma^k Z_S,$$
where
$$C_k^{\mathrm{var}} = 24a_k^k + 128 C_{\mathrm{act},k} C_k^{\mathrm{rec}} < \infty.$$
The constant depends only on fixed k. At $k = 2$ the derivation contains exactly one factor $Z_S = S+1$ and no second factor proportional to S.

Proof. Conditional on c, Proposition A.2 leaves X_i with law D. Integrating Proposition A.6 only after its pointwise level summation gives
$$\begin{aligned}\mathbb{E}[W_i(c)^2 \mid c] &\leq Z_S \left[24h_0^k + 128C_{\mathrm{act},k} \int |x - m_0(c)|^k D(dx)\right] \\ &= Z_S \left[24h_0^k + 128C_{\mathrm{act},k} M_k(c)\right].\end{aligned}$$
On $\mathcal{E}_{\mathrm{loc}}$, Lemma A.2 and $h_0 = a_k\sigma$ imply
$$\begin{aligned}\mathbb{E}[W_i(c)^2 \mid c] &\leq Z_S \left[24a_k^k\sigma^k + 128C_{\mathrm{act},k} C_k^{\mathrm{rec}} \sigma^k\right] \\ &= C_k^{\mathrm{var}} \sigma^k Z_S.\end{aligned}$$
This proves conditional square integrability, and $\mathrm{Var}(W_i(c) \mid c) \leq \mathbb{E}[W_i(c)^2 \mid c]$ gives the variance bound.

At $k = 2$,
$$Z_S = \sum_{s=0}^{S} h_s^0 = S + 1, \qquad p_s = \frac{1}{S+1}, \qquad \frac{h_s^2}{p_s} = (S+1)h_s^2.$$
Consequently Lemma A.12 and the activation ledger, still pointwise in x, give
$$\begin{aligned}\sum_{s=1}^{S} \mathbb{E}[W_i(c)^2 \mathbf{1}\{L_i = s\} \mid X_i = x, c] &\leq 128(S+1) \sum_{s=1}^{S} h_s^2 \mathbf{1}_{\Gamma_s(c)}(x) \\ &\leq 128C_{\mathrm{act},2}(S+1)|x - m_0|^2.\end{aligned}$$
The separate level-zero term is at most
$$24(S+1)h_0^2 \mathbf{1}_{\Gamma_0(c)}(x).$$
After expectation and recentering,
$$\begin{aligned}\mathrm{Var}(W_i(c) \mid c) &\leq (S+1)\left[24h_0^2 + 128C_{\mathrm{act},2} M_2(c)\right] \\ &\leq C_2^{\mathrm{var}} \sigma^2 (S+1) = C_2^{\mathrm{var}} \sigma^2 Z_S.\end{aligned}$$
Thus only the factor arising from p_s^{-1} remains; the pointwise ledger is uniform in S and contributes no second factor.

If $D(J_{0,j_0}) = 1$, Lemma A.10 makes every higher target and alias square exactly zero, leaving only Lemma A.11. If $D\{m_0\} = 1$, that exact formula also vanishes and the conditional variance is zero. These are exact reductions, not remainder bounds.

The separate level-zero and higher-level square identities, their pointwise sum, and the final conditional integration account for every alias square and every importance factor. In particular, the displayed middle-regime calculation proves that the sole level cost is $Z_S = S + 1$. □

A.6 Outer scale, tail control, and three-regime normalization

Lemma A.13 (Admissible dyadic outer scale and exact rounding)**.** *Under Assumption 1 and Lemma A.2, take* $a_k \geq 200$ *and define*

$$b_k = \max\left\{a_k, (4C_k^{\mathrm{rec}})^{1/(k-1)}\right\}.$$

Choose

$$0 < c_k \leq \min\left\{\frac{1}{2}, \left(\frac{b_k}{2a_k}\right)^{k-1}\right\}.$$

For every $0 < \epsilon \leq c_k\sigma$*,*

$$\frac{H_\star}{h_0} \geq 2, \qquad S \geq 1, \qquad H_\star \leq H < 2H_\star.$$

At $H_\star/h_0 = 2$ *one has* $S = 1$ *and* $H = H_\star$*; conversely, under this admissibility condition* $S = 1$ *only at* $H_\star/h_0 = 2$*.*

Proof. The choices depend only on fixed k and give

$$b_k \geq a_k, \qquad b_k^{k-1} \geq 4C_k^{\mathrm{rec}}.$$

Put

$$x = \frac{\sigma}{\epsilon}, \qquad R = \frac{H_\star}{h_0} = \frac{b_k}{a_k}x^{1/(k-1)}.$$

The parameter domain and the choice of c_k imply

$$R \geq \frac{b_k}{a_k}c_k^{-1/(k-1)} \geq \frac{b_k}{a_k}\frac{2a_k}{b_k} = 2.$$

Also $c_k \leq 1/2$ gives $x \geq 2$. Let $q = \log_2 R$. The exact ceiling $S = \lceil q \rceil$ gives $q \leq S < q + 1$, whence

$$H_\star = h_0 R \leq h_0 2^S = H < 2h_0 R = 2H_\star.$$

Since $R \geq 2$, $S \geq 1$. If $R = 2$, then $q = S = 1$ and $H = H_\star$. Conversely $S = 1$ gives $q \leq 1$, while $R \geq 2$ gives $q \geq 1$, so $R = 2$. No strict inequality in the parameter domain was used, and the calculation includes $\epsilon = c_k\sigma$. □

Proposition A.8 (Outer residual at the target scale)**.** *Under Assumptions 1 and 2, Lemmas A.2, A.3, and A.13, and Proposition A.5, every localization transcript in* $\mathcal{E}_{\mathrm{loc}}$ *satisfies*

$$|\mu(D) - m_0 - \theta(c)| \leq \frac{M_k(c)}{H^{k-1}} \leq C_k^{\mathrm{rec}}\frac{\sigma^k}{H^{k-1}} \leq \frac{\epsilon}{4}.$$

If $D(J_{S,j_S}) = 1$*, the left side is exactly zero. This holds in particular when* $D(J_{0,j_0}) = 1$*, preserving the exact level-zero reduction.*

Proof. For a fixed transcript in $\mathcal{E}_{\mathrm{loc}}$, Proposition A.5 gives

$$\mu(D) - m_0 - \theta(c) = \int_{J_{S,j_S}^c} (x - m_0)D(dx).$$

Writing $r = |x - m_0|$, Lemma A.3 gives the strict set inclusion

$$\{x : x \notin J_{S,j_S}\} \subseteq \{x : r > H\},$$

because every point with $r \le H$, including both equality points, lies in the padding. Hence

$$\begin{aligned}|\mu(D) - m_0 - \theta(c)| &\le \int r\mathbf{1}\{x \notin J_{S,j_S}\} D(dx)\\ &\le \int r\mathbf{1}\{r > H\} D(dx)\\ &\le \int \frac{r^k}{H^{k-1}} D(dx) = \frac{M_k(c)}{H^{k-1}}.\end{aligned}$$

The third line uses $r \le r^k/H^{k-1}$ on $r > H$. Lemma A.2 gives the second inequality. Since $H \ge H_\star$ and $k-1 > 0$,

$$\begin{aligned}C_k^{\mathrm{rec}} \frac{\sigma^k}{H^{k-1}} &\le C_k^{\mathrm{rec}} \frac{\sigma^k}{H_\star^{k-1}},\\ H_\star^{k-1} &= b_k^{k-1} \sigma^{k-1} \frac{\sigma}{\epsilon} = b_k^{k-1} \frac{\sigma^k}{\epsilon}.\end{aligned}$$

Thus

$$C_k^{\mathrm{rec}} \frac{\sigma^k}{H^{k-1}} \le C_k^{\mathrm{rec}} b_k^{1-k} \epsilon \le \frac{\epsilon}{4},$$

where the last threshold is exactly $b_k^{k-1} \ge 4C_k^{\mathrm{rec}}$. No tail event is introduced.

The half-open boundary cases are also exact. Since $J_{S,j_S} = [(j_S - 1)H, (j_S + 2)H)$ and $m_0 = (j_S + \vartheta_S)H$ with $0 < \vartheta_S < 1$, an atom at the included left boundary is more than H from m_0 but remains in the truncated integral; an atom at the excluded right boundary is more than H away and is charged by the tail inequality. Atoms at $m_0 \pm H$ lie strictly inside the padding. If $D(J_{S,j_S}) = 1$, the exact residual integral is zero before any inequality. Nesting gives this from $D(J_{0,j_0}) = 1$, and Proposition A.5 then leaves only the exact level-zero mean. □

Lemma A.14 (Exact finite normalizer in the three moment regimes)**.** *Under Assumption 1 and Lemma A.13, write $x = \sigma/\epsilon$. For $k > 2$,*

$$Z_S = h_0^{2-k} \frac{1 - 2^{(2-k)(S+1)}}{1 - 2^{2-k}},$$

and

$$a_k^{2-k} \sigma^{2-k} \le Z_S \le \frac{a_k^{2-k}}{1 - 2^{2-k}} \sigma^{2-k}.$$

For $k = 2$,

$$Z_S = S + 1, \qquad \frac{\log x}{\log 2} \le Z_S < \frac{3 + \log_2(b_2/a_2)}{\log 2} \log x.$$

For $1 < k < 2$,

$$\begin{aligned}Z_S &= h_0^{2-k} \frac{2^{(2-k)(S+1)} - 1}{2^{2-k} - 1},\\ H^{2-k} &\le Z_S \le \frac{H^{2-k}}{1 - 2^{k-2}},\end{aligned}$$

and

$$\begin{aligned}b_k^{2-k} \sigma^{2-k} x^{(2-k)/(k-1)} &\le Z_S\\ &< \frac{2^{2-k} b_k^{2-k}}{1 - 2^{k-2}} \sigma^{2-k} x^{(2-k)/(k-1)}.\end{aligned}$$

Every formula includes the minimum legal case $S = 1$.

Proof. Since $h_s = 2^s h_0$,

$$Z_S = h_0^{2-k} \sum_{s=0}^{S} 2^{(2-k)s}.$$

For $k > 2$, let $\rho = 2^{2-k} \in (0,1)$. Finite multiplication gives $(1-\rho)\sum_{s=0}^{S} \rho^s = 1 - \rho^{S+1}$, proving the exact formula. Moreover,

$$1 \le \sum_{s=0}^{S} \rho^s \le \sum_{s=0}^{\infty} \rho^s = \frac{1}{1-\rho}.$$

Multiplication by $h_0^{2-k} = a_k^{2-k}\sigma^{2-k}$ proves both bounds. The denominator is positive for fixed $k > 2$; no uniform limit as $k \downarrow 2$ is claimed.

At $k = 2$, every term is one. Put

$$R = \frac{H_\star}{h_0} = \frac{b_2}{a_2}x, \qquad q = \log_2 R.$$

The scale lemma gives $b_2/a_2 \ge 1$, $x \ge 2$, and $q \le S < q+1$. Therefore

$$Z_S = S + 1 \ge q + 1 \ge \log_2 x = \frac{\log x}{\log 2}.$$

For the upper bound,

$$\begin{aligned} Z_S &< q + 2 = \log_2 x + \log_2(b_2/a_2) + 2 \\ &\le [3 + \log_2(b_2/a_2)] \log_2 x = \frac{3 + \log_2(b_2/a_2)}{\log 2} \log x. \end{aligned}$$

The absorption is explicit because

$$[3 + \log_2(b_2/a_2)] \log_2 x - [\log_2 x + \log_2(b_2/a_2) + 2] = [2 + \log_2(b_2/a_2)](\log_2 x - 1) \ge 0.$$

For $1 < k < 2$, let $\rho = 2^{2-k} > 1$. The finite identity $\sum_{s=0}^{S} \rho^s = (\rho^{S+1} - 1)/(\rho - 1)$ gives the exact formula. Factoring out the last term yields

$$Z_S = H^{2-k} \sum_{t=0}^{S} 2^{(k-2)t},$$

$$1 \le \sum_{t=0}^{S} 2^{(k-2)t} \le \sum_{t=0}^{\infty} 2^{(k-2)t} = \frac{1}{1 - 2^{k-2}}.$$

Because $2 - k > 0$ and $H_\star \le H < 2H_\star$,

$$H_\star^{2-k} \le H^{2-k} < 2^{2-k} H_\star^{2-k},$$

while

$$H_\star^{2-k} = b_k^{2-k} \sigma^{2-k} x^{(2-k)/(k-1)}.$$

Combining the displays proves the last bounds. The denominator is positive for fixed $1 < k < 2$. When $H_\star/h_0 = 2$, the scale lemma gives $S = 1, H = H_\star$, and the formulas reduce to the two-term sum $h_0^{2-k} + H^{2-k}$, or to 2 at $k = 2$, with no endpoint exception. $\square$

Proposition A.9 (Three-regime refinement scale and nondegeneracy)**.** *Under Assumption 1 and Lemmas A.13 and A.14, define*

$$A_k = \frac{\sigma^k Z_S}{\epsilon^2}, \qquad x = \frac{\sigma}{\epsilon}.$$

For $k > 2$,

$$a_k^{2-k} x^2 \le A_k \le \frac{a_k^{2-k}}{1 - 2^{2-k}} x^2.$$

For $k = 2$,

$$\frac{x^2 \log x}{\log 2} \le A_2 < \frac{3 + \log_2(b_2/a_2)}{\log 2} x^2 \log x.$$

For $1 < k < 2$,

$$b_k^{2-k} x^{k/(k-1)} \le A_k < \frac{2^{2-k} b_k^{2-k}}{1 - 2^{k-2}} x^{k/(k-1)}.$$

Consequently

$$A_k \asymp_k \begin{cases} \sigma^2/\epsilon^2, & k > 2, \\ (\sigma^2/\epsilon^2) \log(\sigma/\epsilon), & k = 2, \\ (\sigma/\epsilon)^{k/(k-1)}, & 1 < k < 2, \end{cases}$$

and A_k has a positive explicit k-only lower bound.

Proof. For $k > 2$, multiplying the normalizer bounds by σ^k/ϵ^2 uses

$$\frac{\sigma^k}{\epsilon^2} \sigma^{2-k} = \frac{\sigma^2}{\epsilon^2} = x^2,$$

and proves the first pair with every geometric factor retained. At $k = 2$, $A_2 = x^2 Z_S$, so multiplying the two exact logarithmic bounds proves the second pair; the additive ceiling term has already been controlled.

For $1 < k < 2$,

$$\begin{aligned} \frac{\sigma^k}{\epsilon^2} \sigma^{2-k} x^{(2-k)/(k-1)} &= x^2 x^{(2-k)/(k-1)} \\ &= x^{2+(2-k)/(k-1)} = x^{k/(k-1)}, \end{aligned}$$

which proves the last pair with its rounding factor and denominator intact.

Finally $x \ge 2$ by Lemma A.13. Thus

$$A_k \ge \underline{A}_k := \begin{cases} 4a_k^{2-k}, & k > 2, \\ 4, & k = 2, \\ 2^{k/(k-1)} b_k^{2-k}, & 1 < k < 2. \end{cases}$$

Every branch is finite, positive, and k-only. All substitutions remain valid at $\epsilon = c_k \sigma$, $S = 1$, and $\lambda = \sigma$. Constants are allowed to deteriorate as a fixed off-middle k approaches two; no continuity through the separately evaluated middle case is asserted.

Together with Proposition A.8, the scale and normalizer lemmas therefore provide the exact $\epsilon/4$ residual certificate, every dyadic boundary case, all three finite-sum formulas, and the positive k-only lower interface for A_k used by the count ceilings. ∎

A.7 Conditional iid refinement and median amplification

Proposition A.10 (Full-transcript conditional iid refinement law)**.** *Under Assumptions 3 and 4, Propositions A.2, A.5, and A.7, conditional on the full localization sigma-field $\mathscr{L}_{\mathrm{loc}}$, the family $(W_i(c))_{i\in I_{\mathrm{ref}}}$ is iid and*

$$\mathbb{E}[W_i(c) \mid \mathscr{L}_{\mathrm{loc}}] = \theta(c) \quad \textit{almost surely.}$$

On $\mathcal{E}_{\mathrm{loc}}$, its common conditional variance satisfies

$$v(c) := \mathrm{Var}(W_i(c) \mid \mathscr{L}_{\mathrm{loc}}) \le C_k^{\mathrm{var}} \sigma^k Z_S \quad \textit{almost surely.}$$

Both moments average over every refinement sample and all refinement seeds; no assertion conditional on frozen refinement coins is made.

Proof. Let Ξ_i denote the complete refinement tuple, assigning fixed dummy values to T_i, B_i on the level-zero branch. Proposition A.2 supplies a version of the conditional law with

$$\mathcal{L}((\Xi_i)_{i\in I_{\mathrm{ref}}} \mid \mathscr{L}_{\mathrm{loc}}) = Q^{\otimes N_{\mathrm{ref}}},$$

where Q is the common refinement sample-and-seed law and is independent of the localization transcript. The statistic is the common measurable map

$$W_i(c) = w(c, \Xi_i),$$

where the transcript enters only through its measurable output c and the resulting decoder path. For a transcript realization ℓ,

$$\mathcal{L}((W_i(c))_{i\in I_{\mathrm{ref}}} \mid \mathscr{L}_{\mathrm{loc}})(\ell) = \bigotimes_{i\in I_{\mathrm{ref}}} \mathcal{L}_Q(w(c(\ell), \Xi_i)).$$

All factors are identical, proving conditional iid under the full transcript.

The conditional mean is

$$\int w(c(\ell), \xi) Q(d\xi) = \theta(c(\ell))$$

by Proposition A.5. On $\mathcal{E}_{\mathrm{loc}}$ the same kernel gives

$$v(c(\ell)) = \int (w(c(\ell), \xi) - \theta(c(\ell)))^2 Q(d\xi),$$

which Proposition A.7 bounds by $C_k^{\mathrm{var}} \sigma^k Z_S$. This equality comes from the explicit transcript-independent kernel and the dependence of w only through c; it does not invoke the generally false claim that enlarging a conditioning sigma-field preserves conditional variance.

If $v(c) = 0$, the displayed integral shows $W_i(c) = \theta(c)$ Q-almost surely. Fixed disjoint groups use disjoint tuple families and are therefore conditionally independent. □

Lemma A.15 (One-group accuracy at the stochastic target)**.** *Under Assumption 1 and Propositions A.10 and A.9, choose*

$$\beta_k \ge 16 C_k^{\mathrm{var}}, \qquad B_{\mathrm{ref}} = \lceil \beta_k A_k \rceil.$$

For every fixed group G_g, almost surely on $\mathcal{E}_{\mathrm{loc}}$,

$$\Pr\left\{ \left| \overline{W}_g(c) - \theta(c) \right| > \frac{\epsilon}{2} \;\middle|\; \mathscr{L}_{\mathrm{loc}} \right\} \le \frac{1}{4}.$$

This includes zero variance and every value of the block ceiling.

Proof. Proposition A.9 gives $A_k > 0$, and the explicit variance constant is positive. Thus the choice is legal and

$$1 \le B_{\mathrm{ref}}, \qquad B_{\mathrm{ref}} \ge \beta_k A_k = \beta_k \frac{\sigma^k Z_S}{\epsilon^2}.$$

Conditional on a transcript in $\mathcal{E}_{\mathrm{loc}}$, Proposition A.10 and the exact group size give

$$\mathbb{E}[\overline{W}_g(c) \mid \mathscr{L}_{\mathrm{loc}}] = \theta(c), \qquad \mathrm{Var}(\overline{W}_g(c) \mid \mathscr{L}_{\mathrm{loc}}) = \frac{v(c)}{B_{\mathrm{ref}}}.$$

Conditional Chebyshev at threshold $\epsilon/2$ yields

$$\begin{aligned}
\Pr\left\{ |\overline{W}_g(c) - \theta(c)| > \frac{\epsilon}{2} \;\middle|\; \mathscr{L}_{\mathrm{loc}} \right\} &\le \frac{4v(c)}{B_{\mathrm{ref}}\epsilon^2} \\
&\le \frac{4C_k^{\mathrm{var}}\sigma^k Z_S}{B_{\mathrm{ref}}\epsilon^2} \\
&= \frac{4C_k^{\mathrm{var}} A_k}{B_{\mathrm{ref}}} \le \frac{4C_k^{\mathrm{var}}}{\beta_k} \le \frac{1}{4}.
\end{aligned}$$

If $v(c) = 0$, every group mean equals $\theta(c)$ conditionally almost surely, and the failure probability is exactly zero. No path or cell union bound is taken. □

Lemma A.16 (Odd-median amplification with exact confidence constants)**.** *Under Assumption 1, Proposition A.10, and Lemma A.15, choose*

$$\alpha_k = 4, \qquad G_\delta = 2\left\lceil 4\log\frac{8}{\delta}\right\rceil + 1.$$

Define the median as the $(G_\delta + 1)/2$-th order statistic, retaining repeated values. Almost surely on $\mathcal{E}_{\mathrm{loc}}$,

$$\begin{aligned}
\Pr\left\{ \left| \operatorname*{median}_{1\le g\le G_\delta} \overline{W}_g(c) - \theta(c) \right| > \frac{\epsilon}{2} \;\middle|\; \mathscr{L}_{\mathrm{loc}} \right\} \\
\le e^{-G_\delta/8} \le \frac{\delta}{2}.
\end{aligned}$$

Proof. Write $G = G_\delta$ and define

$$I_g = \mathbf{1}\left\{ |\overline{W}_g(c) - \theta(c)| > \frac{\epsilon}{2} \right\}, \qquad S_G = \sum_{g=1}^{G} I_g.$$

The fixed disjoint groups are conditionally independent by Proposition A.10; hence the I_g are conditionally independent Bernoulli variables. Lemma A.15 gives, on $\mathcal{E}_{\mathrm{loc}}$,

$$\mathbb{E}[I_g \mid \mathscr{L}_{\mathrm{loc}}] \le \frac{1}{4}, \qquad \mathbb{E}[S_G \mid \mathscr{L}_{\mathrm{loc}}] \le \frac{G}{4}.$$

The chosen G is odd. More generally, write $r = (G+1)/2$ and order the group means as $z_{(1)} \le \cdots \le z_{(G)}$. If $z_{(r)} > \theta(c) + \epsilon/2$, the entries $r, \ldots, G$ give at least r bad groups; if $z_{(r)} < \theta(c) - \epsilon/2$, the entries $1, \ldots, r$ do so. Therefore

$$\left\{ |z_{(r)} - \theta(c)| > \frac{\epsilon}{2} \right\} \subseteq \left\{ S_G \ge \frac{G+1}{2} \right\} \subseteq \left\{ S_G \ge \frac{G}{2} \right\}.$$

An exact tie at either threshold is successful because failure is strict; repeated values do not affect the fixed odd rank.

On the failure event,

$$S_G - \mathbb{E}[S_G \mid \mathscr{L}_{\mathrm{loc}}] \ge \frac{G}{2} - \frac{G}{4} = \frac{G}{4}.$$

Conditional Hoeffding for independent variables in $[0,1]$ gives

$$\begin{aligned}\Pr\left\{|z_{(r)} - \theta(c)| > \frac{\epsilon}{2} \;\middle|\; \mathscr{L}_{\mathrm{loc}}\right\} &\le \Pr\left\{S_G - \mathbb{E}[S_G \mid \mathscr{L}_{\mathrm{loc}}] \ge \frac{G}{4} \;\middle|\; \mathscr{L}_{\mathrm{loc}}\right\} \\ &\le \exp\left(-\frac{2(G/4)^2}{G}\right) = e^{-G/8}.\end{aligned}$$

With $L_\delta = \log(8/\delta) > 0$,

$$G_\delta = 2\lceil 4L_\delta \rceil + 1 \ge 8L_\delta + 1,$$

and hence

$$e^{-G_\delta/8} \le e^{-1/8} e^{-L_\delta} = e^{-1/8}\frac{\delta}{8} < \frac{\delta}{2}.$$

This includes all $\delta \in (0, 1/2)$, including the limiting smallest legal ceiling as $\delta \uparrow 1/2$. If $v(c) = 0$, every I_g is conditionally zero and the median failure probability is exactly zero. $\square$

Proposition A.11 (Conditional refinement accuracy at the common target)**.** *Under Assumption 1, Proposition A.8, and Lemma A.16, define*

$$\mathcal{E}_{\mathrm{ref}} = \left\{ \left| \operatorname*{median}_{1 \le g \le G_\delta} \overline{W}_g(c) - \theta(c) \right| \le \frac{\epsilon}{2} \right\}.$$

Then, almost surely,

$$\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}} \Pr(\mathcal{E}_{\mathrm{ref}}^c \mid \mathscr{L}_{\mathrm{loc}}) \le \frac{\delta}{2} \mathbf{1}_{\mathcal{E}_{\mathrm{loc}}}.$$

On $\mathcal{E}_{\mathrm{loc}} \cap \mathcal{E}_{\mathrm{ref}}$,

$$|\widehat{\mu} - \mu(D)| \le \frac{3\epsilon}{4} < \epsilon.$$

Equivalently,

$$\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}} \Pr\left\{ |\widehat{\mu} - \mu(D)| > \frac{3\epsilon}{4} \;\middle|\; \mathscr{L}_{\mathrm{loc}} \right\} \le \frac{\delta}{2} \mathbf{1}_{\mathcal{E}_{\mathrm{loc}}},$$

and the same inequality holds with threshold ϵ.

Proof. Lemma A.16 gives the first conditional inequality on each successful localization transcript. On the same transcript, Proposition A.8 gives the deterministic bound

$$|\theta(c) - (\mu(D) - m_0)| \le \frac{\epsilon}{4}.$$

On $\mathcal{E}_{\mathrm{ref}}$, the exact estimator definition and the triangle inequality give

$$\begin{aligned}|\widehat{\mu} - \mu(D)| &= \left| \operatorname*{median}_{g} \overline{W}_g(c) - (\mu(D) - m_0) \right| \\ &\le \left| \operatorname*{median}_{g} \overline{W}_g(c) - \theta(c) \right| + |\theta(c) - (\mu(D) - m_0)| \\ &\le \frac{\epsilon}{2} + \frac{\epsilon}{4} = \frac{3\epsilon}{4} < \epsilon.\end{aligned}$$

The stochastic error and deterministic outer residual each appear exactly once. There is no additional localization term because localization only provides the event on which the recentered variance and tail bounds hold. Thus total-error failure on $\mathcal{E}_{\rm loc}$ is contained in $\mathcal{E}_{\rm ref}^c$, proving both indicator-valued inequalities without yet integrating out localization.

If $D(J_{0,j_0(c)}) = 1$, Proposition A.5 makes every higher statistic seedwise zero and sets $\theta(c) = \mu(D) - m_0$, while Proposition A.8 makes the residual zero. The median proof then amplifies only the level-zero correction. If $D\{m_0\} = 1$, every $W_i(c)$, group mean, and median is zero, so $\widehat{\mu} = m_0 = \mu(D)$ exactly.

Proposition A.10, Lemmas A.15 and A.16, and this same-target inequality give the full-localization-transcript conditional interface: failure at most $\delta/2$ and total error $3\epsilon/4 < \epsilon$, including zero variance and the smallest legal odd group count. □

A.8 Query legality, unconditional accuracy, and the rate bridge

Proposition A.12 (Precommitted Borel query certificate)**.** *Under Assumptions 1 and 4, Proposition A.1 and Lemma A.5, every query set used by the two-block protocol is Borel and fixed before the first response. This includes every countable mask realization. The later decoder choice of $c, j_0, (j_s, b_s), m_s, d_s, R_s(c)$ and the coefficients in $W_i(c)$ changes no query and introduces no new query.*

Proof. For localization, Proposition A.1 converts every deterministic source bit into the Borel inverse image $\mathcal{B}_i(R_{\rm loc})$. Its codebook, clipped bins, minimum-index tie rule, sample count, and degenerate seed depend only on known $(\lambda, \sigma, \delta)$ and are fixed before any response. In the zero-query branch $N_{\rm loc} = 0$, so the query family is empty.

For refinement index i, the complete seed

$$(L_i, C_i, T_i, B_i, (\rho_{i,s,j})_{s,j}, U_i),$$

with inapplicable level-zero variables omitted or given dummy values, is drawn before any message. Lemma A.5 proves that the countable sum F_i is jointly Borel, stabilizes pointwise because at most one same-color ring is active, and obeys $|F_i(x)| \le 1$. Hence

$$A_i = \{x \in \mathbb{R} : F_i(x) \ge U_i\}$$

is Borel for every realized seed and dither. Its formula contains the global grid and pre-drawn level, color, type, branch, masks, and dither, but no c, localization bit, selected path, or earlier response.

Only after both complete transcripts have arrived does the decoder evaluate the Borel map $c \mapsto (j_0, j_s, b_s, m_s, d_s, R_s(c))$ and use it to retain already-received bits and form $W_i(c)$. No sample is queried then and no fixed set $\mathcal{B}_i$ or A_i changes. This covers localization failure and atoms at every half-open boundary. □

Proposition A.13 (Exact one-bit communication and fixed horizon)**.** *Under Assumption 4 and Proposition A.12, the protocol has deterministic non-stopping horizon*

$$n = N_{\rm loc} + N_{\rm ref}, \qquad N_{\rm ref} = G_\delta B_{\rm ref}.$$

Each of these independent samples transmits exactly one bit. The baseline $Y_i^0 = \mathbf{1}\{0 \ge U_i\}$ is computed from stored public randomness and transmits no bit; decoder path selection and weighting transmit and query nothing further.

Proof. The localization horizon is selected from known parameters. In the nontrivial branch each $i \in I_{\rm loc}$ sends only

$$Y_i^{\rm loc} = \mathbf{1}\{X_i \in \mathcal{B}_i(R_{\rm loc})\} \in \{0,1\}.$$

In the trivial branch $N_{\text{loc}} = 0$, so no unused sample remains.

The values B_{ref}, G_δ, the set I_{ref}, and its equal group partition are deterministic functions of known parameters and fixed k-only choices. Each $i \in I_{\text{ref}}$ sends only

$$Y_i = \mathbf{1}\{X_i \in A_i\} \in \{0,1\}.$$

The decoder already stores U_i and computes Y_i^0, ΔY_i, the selected path, $W_i(c)$, group means, their median, and $\widehat{\mu}$ locally. Public seeds are not sample messages. Hence every used sample sends one bit and the horizon cannot stop or change with data, messages, seeds, or localization success. □

Proposition A.14 (Unconditional uniform PAC conversion)**.** *Under Assumptions 1– 4, Propositions A.1 and A.11 imply, for every $D \in \mathcal{D}(k, \lambda, \sigma)$,*

$$\Pr_D\{|\widehat{\mu} - \mu(D)| > \epsilon\} \le \frac{\delta}{4} + \frac{\delta}{2} = \frac{3\delta}{4} \le \delta.$$

The probability is unconditional over both sample blocks and every protocol seed, and the inequality is uniform over the unrestricted moment class.

Proof. Fix an admissible D and set $\mathcal{A} = \{|\widehat{\mu} - \mu(D)| > \epsilon\}$. The event $\mathcal{E}_{\text{loc}}$ is $\mathscr{L}_{\text{loc}}$-measurable by Proposition A.2, and Proposition A.11 states almost surely that

$$\mathbf{1}_{\mathcal{E}_{\text{loc}}} \Pr_D(\mathcal{A} \mid \mathscr{L}_{\text{loc}}) \le \frac{\delta}{2} \mathbf{1}_{\mathcal{E}_{\text{loc}}}.$$

The tower property with this measurable multiplier gives

$$\begin{aligned}
\Pr_D(\mathcal{A} \cap \mathcal{E}_{\text{loc}}) &= \mathbb{E}_D[\mathbf{1}_{\mathcal{E}_{\text{loc}}} \mathbf{1}_{\mathcal{A}}] \\
&= \mathbb{E}_D\left[\mathbf{1}_{\mathcal{E}_{\text{loc}}} \Pr_D(\mathcal{A} \mid \mathscr{L}_{\text{loc}})\right] \\
&\le \frac{\delta}{2} \Pr_D(\mathcal{E}_{\text{loc}}) \le \frac{\delta}{2}.
\end{aligned}$$

Proposition A.1 gives $\Pr_D(\mathcal{E}_{\text{loc}}^c) \le \delta/4$. Hence

$$\begin{aligned}
\Pr_D(\mathcal{A}) &\le \Pr_D(\mathcal{E}_{\text{loc}}^c) + \Pr_D(\mathcal{A} \cap \mathcal{E}_{\text{loc}}) \\
&\le \frac{\delta}{4} + \frac{\delta}{2} = \frac{3\delta}{4} \le \delta.
\end{aligned}$$

All remaining refinement samples and seeds are integrated inside the conditional probability. No union bound over cells or paths is used. The estimator is defined on $\mathcal{E}_{\text{loc}}^c$, and that whole event is paid. Since D was arbitrary and every constant is class-uniform, the supremum over $\mathcal{D}(k, \lambda, \sigma)$ follows. □

Proposition A.15 (All-ceiling technical sample bound)**.** *Under Assumption 1, Propositions A.1 and A.9, and Lemmas A.13, A.15, and A.16, there is a finite $\widetilde{C}_k$ depending only on fixed k such that*

$$N_{\text{loc}} + N_{\text{ref}} \le \widetilde{C}_k \left[\log \frac{\lambda}{\sigma} + A_k \log \frac{1}{\delta}\right].$$

The bound includes the dyadic, localization, block, and group ceilings and absorbs $1 + \log(4/\delta)$ through a positive k-only lower bound on A_k.

Proof. Use the complete legal design

$$a_k \geq 200, \qquad b_k = \max\{a_k, (4C_k^{\mathrm{rec}})^{1/(k-1)}\},$$

$$0 < c_k \leq \min\left\{\frac{1}{2}, \left(\frac{b_k}{2a_k}\right)^{k-1}\right\}, \qquad \beta_k = 16C_k^{\mathrm{var}}, \qquad \alpha_k = 4.$$

Every quantity depends only on fixed k. Lemma A.13 proves from the exact ceiling that

$$\frac{H_\star}{h_0} \geq 2, \qquad S \geq 1, \qquad H_\star \leq H < 2H_\star.$$

The threshold $b_k^{k-1} \geq 4C_k^{\mathrm{rec}}$ gives outer bias at most $\epsilon/4$ by Proposition A.8; the choices of β_k and α_k give group failure at most $1/4$ and conditional median failure at most $\delta/2$.

Put

$$L_\delta = \log\frac{1}{\delta}, \qquad \ell_{\lambda,\sigma} = \log\frac{\lambda}{\sigma}.$$

Assumption 1 gives $L_\delta \geq \log 2 > 0$ and $\ell_{\lambda,\sigma} \geq 0$. Proposition A.9 proves

$$A_k \geq \underline{A}_k > 0, \qquad \underline{A}_k = \begin{cases} 4a_k^{2-k}, & k > 2, \\ 4, & k = 2, \\ 2^{k/(k-1)}b_k^{2-k}, & 1 < k < 2. \end{cases}$$

Every branch is positive and depends only on fixed k.

The block ceiling obeys

$$\begin{aligned} \beta_k A_k \leq B_{\mathrm{ref}} = \lceil \beta_k A_k \rceil &< \beta_k A_k + 1 \\ &\leq (\beta_k + \underline{A}_k^{-1})A_k. \end{aligned}$$

Set $C_{B,k} = \beta_k + \underline{A}_k^{-1}$. The odd-group ceiling obeys

$$G_\delta = 2\left\lceil 4\log\frac{8}{\delta}\right\rceil + 1 < 8\log\frac{8}{\delta} + 3.$$

Because

$$\log\frac{8}{\delta} = L_\delta + 3\log 2 \leq 4L_\delta, \qquad 3 \leq \frac{3}{\log 2}L_\delta,$$

we have

$$G_\delta < C_G L_\delta, \qquad C_G = 32 + \frac{3}{\log 2}.$$

Thus both refinement ceilings are retained in

$$N_{\mathrm{ref}} = G_\delta B_{\mathrm{ref}} \leq C_G C_{B,k} A_k L_\delta.$$

Proposition A.1, which already includes the source bin and code-length ceilings, gives

$$N_{\mathrm{loc}} \leq C_{\mathrm{loc},k}[1 + \ell_{\lambda,\sigma} + \log(4/\delta)], \qquad C_{\mathrm{loc},k} = 10001.$$

Furthermore,

$$\log\frac{4}{\delta} = L_\delta + 2\log 2 \leq 3L_\delta, \qquad 1 \leq \frac{L_\delta}{\log 2}.$$

With $C_L = 3 + 1/\log 2$,

$$1 + \log\frac{4}{\delta} \leq C_L L_\delta \leq \frac{C_L}{\underline{A}_k}A_k L_\delta.$$

This is the required localization-confidence absorption. Therefore

$$N_{\rm loc} \le C_{{\rm loc},k}\ell_{\lambda,\sigma} + \frac{C_{{\rm loc},k}C_L}{\underline{A}_k}A_k L_\delta.$$

Adding both blocks yields

$$\begin{aligned} N_{\rm loc} + N_{\rm ref} \le C_{{\rm loc},k}\ell_{\lambda,\sigma} \\ + \left[\frac{C_{{\rm loc},k}C_L}{\underline{A}_k} + C_G C_{B,k}\right] A_k L_\delta. \end{aligned}$$

The proposition follows with

$$\widetilde{C}_k = \max\left\{C_{{\rm loc},k}, \frac{C_{{\rm loc},k}C_L}{\underline{A}_k} + C_G C_{B,k}\right\}.$$

No additive term remains outside $\ell_{\lambda,\sigma} + A_k L_\delta$. □

Proposition A.16 (Rate Specialization Bridge)**.** *Under Assumptions 1– 4, take*

$$a_k \ge 200, \quad b_k = \max\{a_k, (4C_k^{\rm rec})^{1/(k-1)}\}, \quad \beta_k = 16C_k^{\rm var}, \quad \alpha_k = 4,$$

and choose

$$0 < c_k \le \min\left\{\frac{1}{2}, \left(\frac{b_k}{2a_k}\right)^{k-1}\right\}.$$

Then all scale, tail, block, group, confidence, and probability conditions of Propositions A.8, A.11, A.14, and A.15 hold, and there is a finite $C_k > 0$, depending only on fixed k, such that

$$N_{\rm loc} + N_{\rm ref} \le C_k r_k(\lambda, \sigma, \epsilon, \delta).$$

The probability mode is unconditional over all samples and seeds, the horizon is fixed and non-stopping, the norm is absolute value on $\mathbb{R}$, and the hidden constant is independent of $D, \lambda, \sigma, \epsilon, \delta$ and all realized protocol objects.

Proof. The displayed choices are k-only. Lemma A.13 verifies

$$H_\star/h_0 \ge 2, \qquad S \ge 1, \qquad H_\star \le H < 2H_\star,$$

and the choice of b_k verifies the tail threshold

$$C_k^{\rm rec}\frac{\sigma^k}{H^{k-1}} \le \frac{\epsilon}{4}.$$

Lemmas A.15 and A.16 verify the group threshold $\beta_k = 16C_k^{\rm var}$, the odd-group choice $\alpha_k = 4$, conditional group failure at most $1/4$, and conditional median failure at most $e^{-G_\delta/8} \le \delta/2$. Proposition A.11 then gives total conditional error $\epsilon/2 + \epsilon/4 = 3\epsilon/4 < \epsilon$.

The block, group, localization, and confidence ceilings were retained in Proposition A.15. In particular,

$$A_k \ge \underline{A}_k > 0, \quad B_{\rm ref} < (\beta_k + \underline{A}_k^{-1})A_k, \quad G_\delta < \left(32 + \frac{3}{\log 2}\right)\log\frac{1}{\delta},$$

and

$$1 + \log\frac{4}{\delta} \le \frac{3 + 1/\log 2}{\underline{A}_k}A_k \log\frac{1}{\delta}.$$

Consequently

$$N_{\mathrm{loc}} + N_{\mathrm{ref}} \le \widetilde{C}_k \left[\log \frac{\lambda}{\sigma} + A_k \log \frac{1}{\delta}\right].$$

Define

$$Q_k(\sigma, \epsilon) = \begin{cases} \sigma^2/\epsilon^2, & k > 2, \\ (\sigma^2/\epsilon^2)\log(\sigma/\epsilon), & k = 2, \\ (\sigma/\epsilon)^{k/(k-1)}, & 1 < k < 2. \end{cases}$$

Proposition A.9 gives, without asymptotic substitution,

$$A_k \le U_k Q_k(\sigma, \epsilon),$$

where

$$U_k = \begin{cases} \dfrac{a_k^{2-k}}{1 - 2^{2-k}}, & k > 2, \\ \dfrac{3 + \log_2(b_2/a_2)}{\log 2}, & k = 2, \\ \dfrac{2^{2-k} b_k^{2-k}}{1 - 2^{k-2}}, & 1 < k < 2. \end{cases}$$

For each fixed regime the denominator is positive and U_k is finite and k-only. The middle expression comes from the exact identity $Z_S = S + 1$ and contains exactly one factor $\log(\sigma/\epsilon)$. Substitution gives

$$\begin{aligned} N_{\mathrm{loc}} + N_{\mathrm{ref}} &\le \widetilde{C}_k \left[\log \frac{\lambda}{\sigma} + U_k Q_k(\sigma, \epsilon) \log \frac{1}{\delta}\right] \\ &\le \widetilde{C}_k \max\{1, U_k\} \left[\log \frac{\lambda}{\sigma} + Q_k(\sigma, \epsilon) \log \frac{1}{\delta}\right] \\ &= C_k r_k(\lambda, \sigma, \epsilon, \delta), \end{aligned}$$

with $C_k = \widetilde{C}_k \max\{1, U_k\}$. Its dependence is only through fixed-k design and geometric constants.

The probability conversion is also explicit. Proposition A.14 integrates the complete localization sigma-field:

$$\Pr_D(\mathcal{A} \cap \mathcal{E}_{\mathrm{loc}}) = \mathbb{E}_D[\mathbf{1}_{\mathcal{E}_{\mathrm{loc}}} \Pr_D(\mathcal{A} \mid \mathscr{L}_{\mathrm{loc}})] \le \frac{\delta}{2},$$

and then

$$\Pr_D(\mathcal{A}) \le \frac{\delta}{4} + \frac{\delta}{2} = \frac{3\delta}{4} \le \delta.$$

No path union bound appears. The substitutions remain valid at $\lambda = \sigma$, $\epsilon = c_k\sigma$, $S = 1$, and as $\delta \uparrow 1/2$, because $\log(1/\delta) \ge \log 2$ supplies the explicit absorption. No continuity through $k = 2$ is asserted. □

Proposition A.17 (Exact supported-cell and point-mass baselines)**.** *Under Propositions A.3, A.5, A.7, A.8, and A.11, and Lemmas A.4 and A.10, fix any decoder output c. If $D(J_{0,j_0(c)}) = 1$, every retained higher correction is seedwise zero, the higher activation and variance charges vanish, the outer residual is zero, and the estimator is exactly the level-zero unbiased dither correction with the same fixed median aggregation. If $D\{m_0(c)\} = 1$, every $W_i(c) = 0$ seedwise and $\widehat{\mu} = m_0(c) = \mu(D)$.*

Proof. Suppose $D(J_{0,j_0(c)}) = 1$. Path nesting places J_{0,j_0} inside every selected child padding, so every higher target ring is inactive. Four-color separation excludes every retained alias. A mismatched decoder color or branch indicator makes $W_i(c) = 0$ directly. When the indicators match, the sample lies on no retained ring, so $F_i(X_i) = 0$, $Y_i = Y_i^0$, and again $W_i(c) = 0$. This is seedwise equality, not cancellation in expectation.

Thus the higher activation and square ledgers are identically zero. Proposition A.5 reduces to $\theta(c) = \mu(D) - m_0$ using level zero alone. Since $J_{0,j_0} \subseteq J_{S,j_S}$, the outer residual is zero before its moment bound. Hence

$$\widehat{\mu} = m_0 + \operatorname*{median}_{1 \le g \le G_\delta} \left[\frac{1}{B_{\mathrm{ref}}} \sum_{i \in G_g} W_i(c) \mathbf{1}\{L_i = 0\} \right].$$

This is exactly the same level-zero dither estimate and fixed median; the conditional concentration proof adds no higher-ring or tail remainder.

If $D\{m_0(c)\} = 1$, the retained target level-zero amplitude $(X_i - m_0)/(2h_0)$ is zero, while every higher ring is inactive. Whenever a decoder coefficient is nonzero, $F_i(X_i) = 0$ and $\Delta Y_i = 0$. Thus $W_i(c) = 0$ for every sample and every public seed. All group means and the median are zero, and the point-mass condition gives $\widehat{\mu} = m_0(c) = \mu(D)$ with no artificial importance, tail, localization, or confidence residual. ☐

A.9 Proof of the Main Theorem

Proof of Theorem 3.1. Choose $a_k, b_k, c_k, \alpha_k, \beta_k$ as in Proposition A.16. Proposition A.1 provides the always-defined localization output and its fixed cost, while Lemma A.5 and Proposition A.12 prove that all localization and refinement queries are Borel and simultaneously precommitted. Proposition A.13 proves exactly one transmitted bit per used independent sample and the deterministic non-stopping horizon $N_{\mathrm{loc}} + N_{\mathrm{ref}}$.

Proposition A.5 identifies the exact truncated target, Proposition A.7 controls its refinement variance, and Proposition A.8 bounds the sole outer residual by $\epsilon/4$. Proposition A.11 gives conditional error at most $3\epsilon/4 < \epsilon$ with conditional failure at most $\delta/2$. Proposition A.14 integrates the full localization sigma-field and combines this with localization failure $\delta/4$, proving

$$\sup_{D \in \mathcal{D}(k,\lambda,\sigma)} \Pr_D\{|\widehat{\mu} - \mu(D)| > \epsilon\} \le \delta.$$

Proposition A.16 checks every auxiliary threshold and ceiling and proves

$$N_{\mathrm{loc}} + N_{\mathrm{ref}} \le C_k r_k(\lambda, \sigma, \epsilon, \delta)$$

with C_k depending only on fixed k, including exactly one $\log(\sigma/\epsilon)$ in the middle regime. Finally, Proposition A.17 gives the stated supported-cell level-zero reduction and exact point-mass recovery. These conclusions are precisely all clauses of Theorem 3.1. ☐